

Regulation-Induced Balance-Sheet Constraints, Option-Market States, and the Cross-Currency Basis

Abstract

Option-implied volatility and skew systematically condition the balance-sheet costs that underlie deviations from covered interest parity (CIP). These costs are state dependent, jointly influenced by option-market conditions, the dollar cycle, and regulatory capital rules. A stylized model links these states to the slope of dealers' marginal cost function, predicting that volatility widens the basis, while skew and dollar strength amplify the pass-through, which weakens with maturity. Empirical analysis using market data largely confirms these predictions. The results recast CIP deviations as equilibrium prices of intermediation governed by observable option-market and regulatory states, offering investors and dealers a practical state-based signal for hedging, capital budgeting, and macroprudential policy design. The results recast covered interest parity deviations as equilibrium prices of intermediation shaped by observable states, and provide investors and dealers with a practical state-based signal for hedging, capital budgeting, and policy design.

Keywords: Cross-currency basis; covered interest parity; option-implied states; balance-sheet constraints; financial regulation; FX derivatives; dealer intermediation

JEL Classification: F31; G12; G15; G18; G28

1 Introduction

In international finance, the FX options market has become an increasingly vital conduit for transferring and managing currency risk. According to the BIS Triennial Central Bank Survey, average daily turnover in global OTC FX markets reached about \$7.5 trillion in April 2022, with options accounting for roughly 4 percent of that flow. By April 2025, turnover had risen to \$9.6 trillion per day and the share of FX options had nearly doubled to around 7 percent—the fastest expansion among all FX instruments. On the stock side, BIS data indicate that notional amounts outstanding in OTC FX derivatives stood near \$120 trillion by mid-2023, highlighting the market's systemic scale.¹ FX options embed forward-looking information—implied volatility, risk-neutral densities, and risk-reversal skew—that investors and policymakers routinely extract to gauge tail-risk expectations.

¹Bank for International Settlements, *Triennial Central Bank Survey of Foreign Exchange Turnover* (2022, 2025); and *BIS OTC Derivatives Statistics*, June 2023.

Yet, despite their expanding scale and informational richness, the influence of option-implied states on funding costs and cross-currency arbitrage deviations remains formally underexplored. This paper fills that gap by embedding option-market signals within a regulatory-microfounded framework that links Basel capital and margin requirements to dealer intermediation costs in FX forwards and swaps.

Existing research on covered interest parity (CIP) deviations emphasizes macroeconomic flows, dealer leverage, and global dollar funding conditions.² These studies show that when the broad dollar appreciates, cross-currency basis spreads widen and cross-border dollar lending contracts. What is missing is a regulatory-microfounded link that elevates option-implied volatility and skew from ancillary controls to first-order determinants of balance-sheet intermediation costs. Unlike prior studies that include implied volatility or skew as control variables, our framework microfounds their role via capital and margin regulations, deriving sufficient-condition inequalities for volatility-skew-dollar interactions. Our paper provides that link by mapping Basel market-risk rules (FRTB), counterparty-credit exposure charges (SA-CCR), and margining practices (IM/VM) into a dealer cost function that prices FX forwards and swaps.

The stylized model framework delivers four testable predictions which we collectively categorize as propositions. First, higher implied volatility widens the cross-currency basis (more negative). Second, skew amplifies the pass-through of volatility shocks. Third, dollar appreciation further magnifies this amplification, especially at short and intermediate tenors. Fourth, the impact attenuates with maturity, consistent with lower rollover intensity at the long end. Together, these predictions clarify when balance-sheet intermediation becomes most expensive and why the cross-currency basis varies systematically across states and maturities.

We test these mechanisms using data of major currencies across maturities from 3 months to 10 years. Interaction regressions largely confirm that volatility, skew, and the dollar state jointly shape basis dynamics in a state-dependent manner, while regulatory shifts strengthen these channels. Term-structure patterns deliver a taxonomy of which tenors are most exposed to volatility-dollar amplification and which remain insulated.

The results redefine the origins of CIP deviations: not static anomalies or generic funding shocks, but equilibrium prices of intermediation jointly determined by option-market states and regulation. For practice, the contribution is equally direct. A simple state-based indicator comprising implied volatility, skew and the dollar can serve as an early-warning signal of basis widening. This guides hedge-tenor choice, informs IM/KVA budgeting, and helps policymakers design swap-line interventions when private balance-sheet capacity is strained. By unifying option-market signals, regulatory frictions, and the dollar cycle, the paper advances both the academic literature and the practical man-

²See Du, Tepper, and Verdelhan (2018); Avdjiev, Du, Koch, and Shin (2019); Cerutti, Obstfeld, and Zhou (2021).

agement of global dollar funding risk.

The rest of the paper is organized as follows. Section 2 presents the related literature. Section 3 presents the model and propositions. Section 4 describes data and empirical design (baseline, interactions, and regulatory/margin identification. Section 5 outlines the interpretation of results. Section 6 discusses policy implications. Section 7 event study. Section 8 policy implications. Section 9 concludes.

2 Related Literature

Our contribution sits at the intersection of three literatures: (i) exchange rate volatility and its real and financial spillovers, (ii) asset-pricing and options dynamics that extract information from implied volatilities and skews, and (iii) the post-GFC literature on CIP deviations. Across these strands, what is missing is a regulatory-microfounded account that elevates option-implied states—volatility and skew—to first-order determinants of balance-sheet costs in FX intermediation.

A first strand examines exchange-rate volatility and real outcomes. Identification-robust gravity estimates show that average trade-depressing effects of volatility are weaker than earlier reduced-form work (Tenreyro, 2007). Volatility interacts with financial development to shape growth and productivity: it is more damaging in less developed systems (Aghion et al., 2009). These findings highlight that volatility is not mere noise but a systematic state conditioning financing and investment.

A second strand studies volatility and option-implied information in asset markets. Exchange rates, equities, and flows co-move through portfolio-balance channels (Hau & Rey, 2006), and firm-level FX exposure is priced in stock returns (Jorion, 1990). FX options provide tail-sensitive hedges and forward-looking signals. The Garman–Kohlhagen model underpins the extraction of implied volatilities (Garman & Kohlhagen, 1983), and option-implied volatility has been shown to transmit across currency and asset markets (Bubák et al., 2011). Risk-reversal skew captures asymmetry and crash risk; demand-based pricing shows option surfaces shift with imbalances (Bollen & Whaley, 2004; Gârleanu et al., 2009). In FX specifically, stochastic skew improves pricing and risk management (Carr & Wu, 2007). Macroeconomic uncertainty is an additional driver: policy uncertainty indices explain variation in implied volatility across countries and time (Krol, 2014; Bartsch, 2019). Together, this literature treats implied volatility and skew as information aggregators, but not yet as determinants of intermediation supply costs.

Recent advances in intermediary-based and international asset pricing further contextualize these dynamics. He & Krishnamurthy (2013) and Adrian, Etula & Muir (2014) establish the theoretical and empirical foundation for intermediary asset pricing, linking balance-sheet capacity to risk premia across markets. He et al. (2017) extend this frame-

work to a multi-asset setting, showing that intermediaries’ marginal value of wealth prices a wide array of returns. Gabaix & Maggiori (2015) and Miranda-Agrippino & Rey (2020) highlight the dominant role of the U.S. dollar in global financial cycles, implying that dollar funding tightness operates as a global constraint shaping cross-border pricing relationships. Complementary evidence by Maggiori, Neiman & Schreger (2020) shows how currency hierarchy influences international capital allocation and reinforces the dollar’s systemic role. On the option-market side, Jurek (2014) and Andersen, Fusari & Todorov (2015) analyze risk premia embedded in currency and equity options, demonstrating how volatility and skew dynamics embed compensation for crash and funding risk. Mancini, Ranaldo & Wrampelmeyer (2013) provide micro-level evidence that FX liquidity risk is priced and co-moves with volatility, consistent with balance-sheet channels of transmission. Within market infrastructure, Duffie & Zhu (2011) clarify how central clearing and initial margining reshape counterparty exposures and liquidity usage—mechanisms directly relevant to the cost slope in our model. Finally, Koijen & Yogo (2019) formalize how institutional demand pressures drive equilibrium prices, offering a complementary framework to our interpretation of hedging demand and dealer intermediation.

A third strand links volatility and tail risk to currency premia. Carry returns load on crash risk and funding liquidity (Brunnermeier et al., 2009), and common factors in currency returns reflect exposure to global risk (Lustig et al., 2011). High-volatility states compress carry premia and tighten intermediary constraints (Menkhoff et al., 2012). Parallel theories of margin and funding constraints explain how volatility can steepen effective supply elasticities: funding and market liquidity interact in “liquidity spirals” (Brunnermeier & Pedersen, 2009), while tighter margins increase required returns and generate deviations from the law of one price (Gârleanu & Pedersen, 2011). These mechanisms rationalize why volatility shocks are amplified through balance-sheet constraints.

The post-crisis CIP literature establishes persistent deviations as a central fact. Du et al. (2018) document their size and persistence across currencies and maturities, rejecting explanations based purely on credit risk. Avdjiev et al. (2019) identify the triangle: dollar appreciation coincides with wider bases and slower cross-border lending, consistent with tighter balance-sheet capacity. At the macro-financial level, Cerutti et al. (2021) relate CIP deviations to global financial conditions and country characteristics. This literature establishes which states correlate with deviations but remains largely silent or reduced form on the role of option-market states.

To sum up, prior work shows that (i) exchange-rate volatility is macro- and finance-relevant, (ii) options embed forward-looking information and reveal demand-supply pressures, and (iii) CIP deviations reflect balance-sheet constraints. What is missing is a regulatory-microfounded bridge: a framework that maps option-implied volatility and skew into dealer marginal costs via FRTB, SA-CCR, and margin mechanics. Our paper provides that bridge, showing how option states become important considerations that

govern the shadow cost of intermediation and hence the dynamics of the cross-currency basis.

3 Regulation: Basel Capital and Margin Frameworks

Before presenting the stylized model, we summarize the regulatory foundation that governs the cost of FX intermediation. Three channels—market-risk capital, counterparty-credit exposure, and margin requirements—jointly determine the effective slope of dealer marginal cost. These channels map directly into the valuation adjustments desks internalize and ultimately into the cross-currency basis.

3.1 Market-Risk Capital (FRTB)

The **Fundamental Review of the Trading Book (FRTB)** sets Basel standards for trading-book capital (including FX forwards, swaps, and options). Dealers must either (i) use the *Standardized Approach* with sensitivities-based charges for delta, vega, and curvature, plus a default-risk charge, or (ii) qualify for an *Internal Models Approach* based on expected shortfall at the 97.5% level with liquidity horizons and backtesting. In both cases, higher implied volatility or tail asymmetry (skew) raises charges on vega/curvature risk, so required market-risk capital rises for a given inventory, tightening balance-sheet capacity.

3.2 Counterparty Credit Exposure (SA-CCR)

The **Standardised Approach for Counterparty Credit Risk (SA-CCR)** converts a netting set into an exposure-at-default (EAD),

$$\text{EAD} = \alpha (\text{RC} + \text{PFE}), \quad \alpha \approx 1.4.$$

Here $\text{RC} = \max(V - C, 0)$ is replacement cost (MTM less collateral) and PFE is a standardized add-on that rises with notional, maturity, hedging set, and option deltas. EAD feeds directly into risk-weighted assets (RWA) and hence equity capital requirements. In volatile states, larger MTM swings and option effective notionals raise both RC and PFE, so credit capital intensifies. V is the current mark-to-market (MTM) value of the derivative portfolio in the bank's favor, C is the value of collateral held against that portfolio, and the term PFE stands for *Potential Future Exposure*, a standardized add-on that rises with notional size, maturity, the assigned hedging set, and option deltas (via supervisory formulas that map each contract into an effective notional).

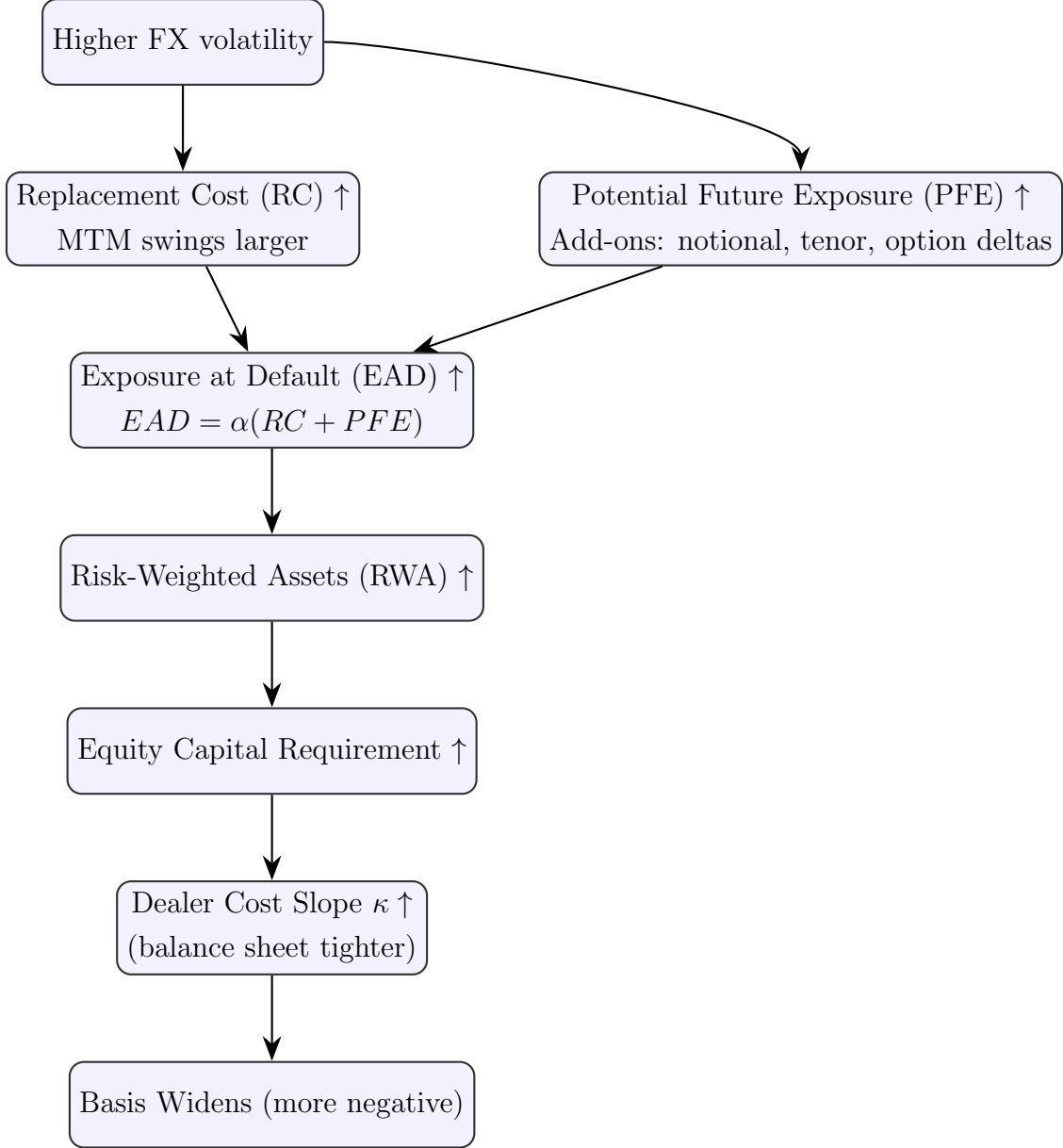


Figure 1: SA-CCR transmission: volatility $\rightarrow$ RC/PFE $\rightarrow$ higher EAD $\rightarrow$ higher RWA $\rightarrow$ higher capital $\rightarrow$ higher cost slope κ $\rightarrow$ wider basis.

3.3 Margins and Liquidity (UMR and CCP)

Collateral rules add a liquidity channel. Variation margin (VM) covers daily MTM moves; initial margin (IM) covers potential moves over a margin period of risk. Under the **Uncleared Margin Rules (UMR)**, most non-cleared derivatives post bilateral IM/VM, with a key carve-out for physically-settled FX forwards and swaps (IM-exempt). FX options and NDFs, however, are in-scope. For cleared products, CCPs set IM via VaR/ES models and call VM daily (often intraday). Empirically, both IM and VM flows surge in volatility spikes, creating procyclical liquidity drains (e.g., March 2020).

Table 1: SA-CCR vs. Margin: When Capital vs. Liquidity Bites

Case	Condition	RC	VM	Balance-sheet effect	Effect on κ
Dealer in-the-money	$V - C > 0$	$RC > 0$	Inflow	Higher $EAD \Rightarrow RWA \uparrow \Rightarrow$ capital need $\uparrow$	$KVA \uparrow \Rightarrow \kappa \uparrow$
Dealer out-of-the-money	$V - C < 0$	$RC = 0$	Outflow	Liquidity drain from VM (and IM if in-scope)	$MVA \uparrow \Rightarrow \kappa \uparrow$
Volatility shock $\uparrow$	(either case)	raise $ V - C $	raise VM freq/size	EAD (via RC/PFE) and/or cash needs rise	

V = MTM value of exposure; C = collateral provided. $RC = \max(V - C, 0)$ replace cost. PFE = Potential Future Exposure. $EAD = \alpha(RC + PFE)$ feeds credit RWAs and capital—priced via KVA . VM/IM are funded liquidity outlays—priced via MVA . When $V - C > 0$, capital needs dominate ($\Rightarrow KVA \uparrow$). When $V - C < 0$, cash outflows dominate ($\Rightarrow MVA \uparrow$). Both steepen κ , widen basis.

3.4 KVA, MVA, and the Cost Slope

Desks internalize these channels through valuation adjustments. The *capital valuation adjustment* (KVA) prices the return required on equity held against FRTB and SA-CCR RWAs. The *margin valuation adjustment* (MVA) prices the funding cost of IM and liquidity for VM:

$$KVA \approx \text{Required capital} \times \text{hurdle ROE}, \quad MVA \approx \text{PV of IM funding cost}.$$

Both rise when implied volatility (σ) or skew (RR) increase because volatility and skew steepen capital and margin requirements.

4 Model

Having described relevant building blocks for the microfounded regulatory framework, we now model the cross-currency basis as the price of the dealer balance sheet required to intermediate FX forwards/swaps. Two important points guide the setup. First, when the broad USD appreciates, the basis widens and cross-border dollar lending slows, Avdjiev et al. (2019), which we take as evidence that the USD, at least at some levels, indexes the shadow price of risk-taking capacity. Second, implied FX volatility and tail asymmetry (risk-reversal skew) raise warehousing costs for short-gamma/vega dealers via capital, exposure, and margin channels. These observations motivate a state-dependent marginal cost slope that loads on the dollar state, implied vol, skew, and tenor.

End-users (asset managers, banks, and corporates) have downward-sloping local inverse demand for intermediation; dealers supply notional subject to a convex, state-dependent balance-sheet cost. The local linear price–quantity mapping we use later, $p_{i,t} = a_i - b_i q_{i,t}$, is the standard workhorse in liquidity/intermediation: prices move

linearly with order flow in Kyle (1985), clientele demand is linear in preferred-habitat term-structure models (Vayanos–Vila, 2021), and option prices move proportionally to demand pressure in demand-based option pricing.³

Our regulatory mapping is such that state variables enter the cost slope through observable rules. (i) *Market risk capital (FRTB)*—vega/curvature sensitivities increase with implied vol and skew. (ii) *Counterparty credit (SA-CCR)*—exposure-at-default depends on replacement cost (RC) and potential future exposure (PFE), which rise with volatility. (iii) *Margining (IM/VM)*—initial and variation margins are procyclical with volatility; March 2020 saw a spike in CCP VM and higher IM, directly raising funding needs. These channels justify loading the cost slope on σ and RR and explain stronger pass-through in stress.⁴

Conventions and timing. We measure the basis $x_{i,t}(T)$ vis-à-vis USD (more negative x = wider basis). The dollar state D_t is the FRB trade-weighted broad USD index. Option states are ATM implied volatility $\sigma_{i,t}$ (e.g., 3M) and the 25Δ *signed* risk-reversal $RR_{i,t}$. The mapping $x = -p$ reflects that the dealer’s cash-flow price is the negative of the basis under standard market quoting.

It should be noted that Risk-Weighted Assets (RWA) are not the raw asset total; they are the exposure scaled by regulatory risk weights (credit) or by modelled tail risk capital (market). Thus higher EAD or higher ES raises RWA—and hence capital—even if the balance-sheet asset amount is unchanged. Below, we now explain the model.

4.1 Environment

A representative dealer supplies USD funding via FX forwards/swaps. Let $x_{i,t}(T)$ be the cross-currency basis for currency i and tenor T , where more negative x means a wider basis. Let D_t denote the broad USD index; $\sigma_{i,t}$ the ATM implied volatility; and $RR_{i,t}$ the 25Δ risk-reversal.

4.2 End-user demand

Following standard liquidity and intermediation models, we adopt a local linear inverse-demand approximation for intermediation services,

$$p_{i,t} = a_i - b_i q_{i,t}, \quad a_i, b_i > 0, \quad (1)$$

³Kyle (1985, *Econometrica*); Vayanos and Vila (2021, *Econometrica*); Garman–Kohlhagen (1983) for FX option valuation and Greeks underpinning the demand-based price sensitivity.

⁴Basel FRTB (2019) minimum capital requirements; Basel Framework CRE52 (SA-CCR); BCBS–CPMI–IOSCO (2022) *Review of Margining Practices* documenting VM peaks and IM responsiveness.

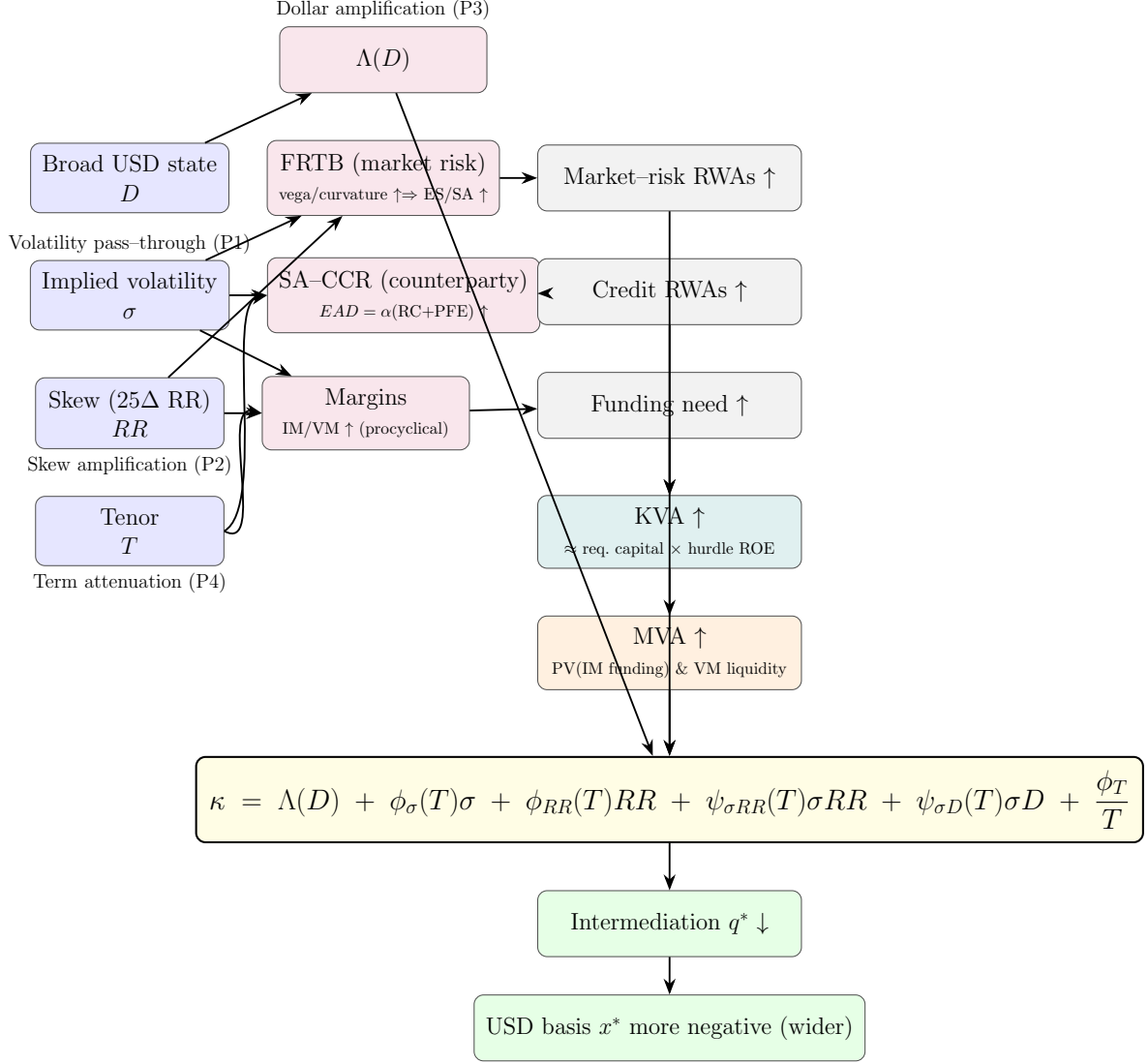


Figure 2: Vol/skew $\rightarrow$ FRTB/SA—CCR/margins $\rightarrow$ KVA/MVA $\rightarrow$ higher slope $\kappa \rightarrow$ less q^* and a wider (more negative) USD basis.

where $q_{i,t}$ is the notional supplied by the dealer and $p_{i,t} > 0$ is the cash-flow price paid to the dealer (we map the basis as $x = -p$). Linear price–quantity mappings are workhorses in market microstructure and preferred-habitat settings as in Kyle’s (1985) model prices move linearly with order flow; in Vayanos and Vila (2021, Eq. (5)) clientele demand is linear and decreasing in (log) price with a maturity-specific slope; and in Gârleanu, Pedersen, and Poteshman (2009) option prices move proportionally to demand pressure, providing a local linear demand–price relation around the current state.⁵

⁵Kyle (1985, *Econometrica*); Vayanos and Vila (2021, *Econometrica*); Gârleanu, Pedersen, and Poteshman (2009, *RFS*). See also Greenwood and Vayanos (2014, *RFS*) and Krishnamurthy and Vissing-Jorgensen (2012, *JPE*) for linear demand/supply effects in fixed-income.

4.3 Dealer cost

We assume dealers are competitive price takers with convex costs. A dealer chooses q to maximize $\pi = pq - C(q)$, so the first-order condition is

$$p = C'(q) \quad (\text{price equals marginal cost}).$$

with a quadratic cost,

$$C_{i,t}(q) = \frac{1}{2} \kappa_{i,t}(T) q^2 \Rightarrow p = \kappa_{i,t}(T) q,$$

which provides a linear inverse supply with slope $\kappa_{i,t}(T)$.⁶

State-dependent slope.

Intermediation capacity tightens when funding and margin constraints bind, so the (price–quantity) supply slope is higher in stressed states. In liquidity models, higher volatility raises margins and tightens funding, increasing price impact (i.e. a liquidity spiral); margin-based asset pricing links wider bases to higher shadow costs of capital.⁷

Regulatory microfoundations. Under the Basel market-risk standard (FRTB), vega and curvature sensitivities increase capital when implied volatility and curvature rise; under SA-CCR, exposure-at-default (EAD) depends on replacement cost and potential future exposure, which rise with volatility/tenor; and initial/variation margins are procyclical in volatility (as in March 2020), all of which elevate the shadow price of the balance sheet.⁸

FX states and the cost slope. We parameterize the marginal-cost slope as a function of observable states with economically motivated interactions:

$$\begin{aligned} \kappa_{i,t}(T) = & \underbrace{\Lambda(D_t)}_{\text{USD state}} + \underbrace{\phi_\sigma(T) \sigma_{i,t}}_{\text{volatility load (FRTB vega/curv., SA-CCR EAD, IM/VM)}} \\ & + \underbrace{\phi_{RR}(T) RR_{i,t}}_{\text{signed tail asymmetry}} + \underbrace{\psi_{\sigma RR}(T) \sigma_{i,t} RR_{i,t} + \psi_{\sigma D}(T) \sigma_{i,t} D_t}_{\text{state-dependent amplification}} \\ & + \underbrace{\frac{\phi_T}{T}}_{\text{tenor}}, \quad \Lambda'(D_t) > 0. \end{aligned} \tag{2}$$

⁶In perfect competition a firm takes p as given and chooses q so that $p = C'(q)$; the marginal-cost curve is the (inverse) supply. See, e.g., OpenStax *Principles of Economics* Ch. 8 (“In a perfectly competitive market, price will be equal to marginal cost”), and CORE *The Economy* 2.0, §8.4 (“a price-taking firm accepts the market price and chooses q so that $MC = p$ ”).

⁷Brunnermeier and Pedersen, “Market Liquidity and Funding Liquidity”; Gârleanu and Pedersen, “Margin-Based Asset Pricing and Deviations from the Law of One Price”.

⁸Basel Committee (2019) *Minimum capital requirements for market risk* (FRTB); Basel Framework CRE52 (SA-CCR); BCBS–CPMI–IOSCO (2022) *Review of margining practices*.

(i) A stronger broad USD tightens global risk-taking capacity, so $\Lambda'(D_t) > 0$.⁹ (ii) Higher implied FX volatility raises vega/curvature capital (FRTB), increases exposure-at-default (SA-CCR), and lifts IM/VM, so the volatility loading $\phi_\sigma(T)$ is positive.¹⁰ (iii) The risk-reversal $RR_{i,t}$ loads via $\phi_{RR}(T)$, and volatility-skew and volatility-USD interactions enter through $\psi_{\sigma RR}(T)$ and $\psi_{\sigma D}(T) \geq 0$, reflecting that vol shocks bite more when tails are skewed one-sided (a rise in risk reversal) or when the dollar has appreciated. (iv) Tenor enters via ϕ_T/T and through the tenor dependence of the loads $\phi_\sigma(T), \phi_{RR}(T), \psi_{\sigma RR}(T), \psi_{\sigma D}(T)$, which decline with T . We can interpret ϕ_T as the *per-roll* unit cost of re-establishing short-dated hedges, so the annualized contribution scales with roll frequency ($\approx 1/T$). For example, $T=0.25\text{year}$ (3M) $\Rightarrow$ 4 rolls/year, whereas $T=5\text{y} \Rightarrow 0.2$ rolls/year.

4.4 Equilibrium and mapping to the basis

Market clearing under competition. End-users face inverse demand $p_{i,t} = a_i - b_i q_{i,t}$; competitive dealers choose q so that $p = C''(q)$, and with $C_{i,t}(q) = \frac{1}{2} \kappa_{i,t}(T) q^2$ this implies a linear inverse supply $p = \kappa_{i,t}(T) q$. Market clearing gives

$$a_i - b_i q_{i,t} = \kappa_{i,t}(T) q_{i,t} \quad \Rightarrow \quad q_{i,t}^* = \frac{a_i}{b_i + \kappa_{i,t}(T)}, \quad p_{i,t}^* = \frac{\kappa_{i,t}(T) a_i}{b_i + \kappa_{i,t}(T)}.$$

Mapping to the basis. By market convention the cross-currency basis equals the negative of the dealer cash-flow price,¹¹ so $x_{i,t}^* = -p_{i,t}^*$ and

$$x_{i,t}^*(T) = -\frac{\kappa_{i,t}(T) a_i}{b_i + \kappa_{i,t}(T)}. \quad (3)$$

A higher cost slope $\kappa_{i,t}(T)$ (tighter balance-sheet state) reduces equilibrium inter-mediation q^* and makes the USD basis more negative: $\frac{\partial q^*}{\partial \kappa} = -\frac{a_i}{(b_i + \kappa)^2} < 0$ and $\frac{\partial x^*}{\partial \kappa} = -\frac{a_i b_i}{(b_i + \kappa)^2} < 0$. This links any state variable that rises κ (e.g., stronger USD, higher implied vol, larger skew, shorter tenor) to a wider (more negative) USD basis.

⁹Avdjiev, Du, Koch, and Shin (2019, *AER Insights*); BIS WP 592.

¹⁰Basel Committee (2019) *Minimum capital requirements for market risk* (FRTB); Basel Framework CRE52 (SA-CCR); BCBS-CPMI-IOSCO (2022) *Review of margining practices*.

¹¹The basis is the cash-market USD rate minus the synthetic USD rate from the swap; a more negative USD basis indicates costlier USD funding via FX swaps. See Borio et al. (2016, *BIS Quarterly Review*); Du, Tepper, and Verdelhan (2018). By market convention, the basis is defined such that it equals the negative of the dealer's present value adjustment per unit notional, i.e. $x_{i,t}^* = -p_{i,t}^*$ where $x_{i,t}^*$ is the market-quoted basis, and $p_{i,t}^*$ is the present value of the dealer's basis-related cash-flow adjustment.

4.5 Propositions

The equilibrium basis in (3) is $x_{i,t}^*(T) = -\kappa_{i,t}(T)a_i/(b_i + \kappa_{i,t}(T))$. For compactness define

$$x_k \equiv \frac{\partial x^*}{\partial \kappa} = -\frac{a_i b_i}{(b_i + \kappa)^2} < 0, \quad x_{kk} \equiv \frac{\partial^2 x^*}{\partial \kappa^2} = \frac{2a_i b_i}{(b_i + \kappa)^3} > 0.$$

Hence any state that raises the cost slope $\kappa_{i,t}(T)$ —through the USD state, implied volatility, tail asymmetry, or tenor—reduces the equilibrium quantity q^* and makes the cross-currency basis more negative. The comparative statics below formalize four testable predictions and link them to microfounded cost channels (FRTB vega/curvature capital, SA-CCR exposure-at-default, and IM/VM margining) and to the broad dollar.¹²

Proposition 1: P1 (Volatility pass-through).

$$\frac{\partial x^*}{\partial \sigma_{i,t}} = x_k \left(\phi_\sigma(T) + \psi_{\sigma RR}(T) RR_{i,t} + \psi_{\sigma D}(T) D_t \right) < 0, \quad \phi_\sigma(T) + \psi_{\sigma RR}(T) RR_{i,t} + \psi_{\sigma D}(T) D_t > 0.$$

Economic interpretation. Higher implied FX volatility increases the cost of supplying hedges for short-gamma/vega dealers along FRTB, SA-CCR, and margin channels; the effective vol load is larger when skew is one-sided or the USD state is tight. Indeed, higher implied FX volatility increases the cost of supplying hedges for short-gamma/vega dealers along three observable channels: (i) *market-risk capital* rises via FRTB’s vega/curvature sensitivities; (ii) *counterparty exposure* rises under SA-CCR through PFE add-ons and replacement-cost effects; and (iii) *margins* (IM/VM) scale up procyclically with volatility, as documented in March 2020. These tighten the effective supply elasticity (raise κ), so a given demand shock translates into a more negative basis.¹³

Proposition 2: P2 (Skew amplification).

$$\frac{\partial^2 x^*}{\partial \sigma_{i,t} \partial RR_{i,t}} = x_{kk} \left(\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D \right) \left(\phi_{RR} + \psi_{\sigma RR} \sigma \right) + x_k \psi_{\sigma RR}(T).$$

Economic interpretation. The term $x_k \psi_{\sigma RR}(T)$ is *negative* if $\psi_{\sigma RR}(T) > 0$ and dominates and delivers state-contingent amplification: vol shocks widen the basis more when skew is large and of the adverse sign for dealers. Indeed, when the 25 Δ risk reversal (skew) is large, one tail of the distribution is especially expensive. A short-vega dealer then warehouses risk precisely in that costly wing when clients request option protection in that

¹²Funding/margin constraints tighten supply slopes (Brunnermeier and Pedersen, 2009); margins and funding costs widen bases (Gârleanu and Pedersen, 2011). Regulatory channels: FRTB (Basel, 2019) loads capital on vega/curvature; SA-CCR (CRE52) defines EAD=RC+PFE; March 2020 saw outsized VM/IM spikes (BCBS-CPMI-IOSCO, 2022); the USD “triangle” links a strong broad USD to wider USD bases and slower cross-border USD lending (Avdjiev, Du, Koch, and Shin, 2019; BIS WP 592).

¹³FRTB: Basel Committee (2019), minimum capital requirements for market risk; SA-CCR: Basel Framework CRE52; margin spikes: BCBS-CPMI-IOSCO (2022) (e.g., CCP VM peak $\sim$ \$140bn on 9 Mar 2020). See also Brunnermeier and Pedersen (2009) on volatility-margin feedbacks.

extreme wing. Volatility shocks raise option values most in that wing, amplifying losses, and in turn raise vega/curvature capital charges (FRTB) and IM/VM calls (margining). The result is that balance-sheet costs increase *more than proportionally*, steepening the supply curve of intermediation. Hence the volatility-to-basis pass-through is stronger (more negative) when skew is high.

Proposition 3: P3 (Dollar-state amplification).

$$\frac{\partial^2 x^*}{\partial \sigma_{i,t} \partial D_t} = x_{kk} \left(\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D \right) \left(\Lambda'(D_t) + \psi_{\sigma D} \sigma \right) + x_k \psi_{\sigma D}(T).$$

Again, the interaction term $x_k \psi_{\sigma D}(T)$ is *negative* if $\psi_{\sigma D}(T) > 0$ and dominates, so volatility hurts more in strong-dollar (tight capacity) states.

Economic interpretation. A stronger broad USD is a barometer of tighter global balance-sheet capacity. When D_t is high, the shadow price of balance sheets $\Lambda(D_t)$ is elevated; a volatility shock then raises the slope κ from an already tight level, producing a larger widening of the basis. Thus, states that strengthen the USD also magnify volatility pass-through to the basis.

Proposition 4: P4 (Term-structure attenuation). Since both the denominator $(b_i + \kappa)^{-2}$ and the loads depend on T ,

$$\left| \frac{\partial x^*}{\partial \sigma_{i,t}} \right| = \frac{a_i b_i}{(b_i + \kappa)^2} \left| \phi_\sigma(T) + \psi_{\sigma RR}(T) RR + \psi_{\sigma D}(T) D \right|.$$

Empirically, attenuation with T arises because the *loads* $\phi_\sigma(T), \psi_{\sigma RR}(T), \psi_{\sigma D}(T)$ decline with maturity (roll frequency, near-dated PFE/margin sensitivity, and capital multipliers are front-loaded), which dominates the mechanical $(b_i + \kappa)^{-2}$ term.

Economic interpretation. Short-tenor intermediation rolls more frequently and faces short-horizon margining and PFE add-ons, so front-end basis quotes (e.g., 3M) are more state-sensitive than long-tenor quotes (e.g., 5Y) when the *loads* $\phi_\sigma(T), \psi_{\sigma RR}(T), \psi_{\sigma D}(T)$ decline with maturity (roll frequency, near-dated PFE/margin sensitivity, and capital multipliers are front-loaded), which dominates the mechanical $(b_i + \kappa)^{-2}$ term.

5 Implied Volatility and the Cross-Currency Basis

This section takes the model's propositions to the data and examines how implied FX volatility relates to the cross-currency basis across tenors (3M, 1Y, 5Y, 10Y). After presenting summary statistics for implied volatility and the basis, we document (i) a strong contemporaneous relationship between implied volatility and CIP deviations (ii) amplification when the broad USD is strong and when option skew (risk-reversal) is large; and (iii) reduction of the volatility effect at longer maturities, consistent with less frequent

hedge rebalancing and lower roll-driven balance-sheet costs.

5.1 Summary Statistics

Our empirical analysis focuses on liquid USD pairs, subject to data availability: AUD, CHF, EUR, GBP, JPY, and NOK. These currencies account for more than half of global daily FX turnover. The main measures are:

- **Basis** $x_{i,t}(T)$: USD cash rate minus the USD rate implied by swapping currency i into USD at tenor T (more negative, $x \Rightarrow$ wider basis).
- **Implied volatility** $\sigma_{i,t}$: ATM IV (e.g., 3M).
- **Skew** $RR_{i,t}$: 25 Δ risk-reversal. (call IV – put IV)
- **Dollar state** D_t : FRB trade-weighted broad USD index.

We note that the skew, i.e., risk reversal, variables do not always have the same base and quote currency. For EUR, GBP and AUD, the quote currency is the USD, whereas for JPY, CHF and NOK, the base currency is the USD. To approximate these measures so that all of them have the same quote currency, we multiply the risk reversals of all currencies with a USD base by -1. This way, we express all the risk reversals in the same quote currency, the USD. This would then imply that for all the currencies’ risk reversals, an increase or a positive change would signify that call IV rises faster than put IV, which implies the market’s expectation of an appreciation of the base currency or, alternatively, a depreciation of the USD.

Based on data availability, we use weekly data for all the tenors we consider (3M, 1Y, 5Y, 10Y). All raw series are from Bloomberg. Our sample spans January 2007 to December 2021, but coverage differs because some currencies start much later.

Table 2: Implied volatility summary (% , sample)

	AUD	JPY	CHF	EUR	GBP	NOK
Mean	11.00	8.65	9.23	9.08	9.24	11.53
s.d.	3.86	2.22	2.90	3.18	2.98	3.36
CV (%)	35.1	30.5	31.4	35.0	32.3	29.1

Table 1 reports implied volatilities for six USD currency pairs. Average levels are highest for NOK (11.53) and AUD (11.00), while JPY (8.65), CHF (9.23), EUR (9.08), and GBP (9.24) are lower. Standard deviations range from 2.90 (CHF) to 3.86 (AUD). The coefficient of variation (CV)—a measure of relative dispersion—is highest for AUD (35.1%) and EUR (35.0%), and lowest for NOK (29.1%), indicating moderately elevated relative volatility for AUD and EUR versus the others. These patterns suggest that

Table 3: Cross-currency basis spreads (bps): summary statistics by currency and tenor

Currency	Mean (bps)				S.d. (bps)			
	3M	1Y	5Y	10Y	3M	1Y	5Y	10Y
EUR (EU)	-26.92	-21.44	-20.21	-17.33	25.32	19.05	15.72	13.75
JPY (JY)	-26.23	-26.52	-42.32	-42.69	18.07	18.49	27.95	27.89
GBP (BP)	-13.47	-8.44	-5.83	-5.68	20.64	12.35	11.17	10.45
AUD (AD)	9.51	10.84	19.24	21.14	11.92	8.54	9.90	14.68
CHF (SF)	-27.07	-19.50	-24.93	-28.42	20.79	15.51	17.73	19.51
NOK (NK)	—	-20.34	-11.69	-10.00	—	16.42	8.89	6.88

Notes: Spreads in basis points (bps). “—” indicates 3M statistics not available in the provided data. Currency tags in parentheses follow the series codes: EU=EUR, JY=JPY, BP=GBP, AD=AUD, SF=CHF, NK=NOK.

option-implied uncertainty is comparatively greater for AUD and NOK, consistent with commodity- and carry-sensitive currencies, while EUR, JPY, CHF, and GBP exhibit lower and more stable implied volatilities over the sample.

Table 2 summarizes cross-currency basis swap spreads (basis points) for 3M, 1Y, 5Y, and 10Y (3M unavailable for NOK). At 1Y, AUD shows a positive basis (+10.84 bps), while EUR (−21.44 bps), JPY (−26.52 bps), CHF (−19.50 bps), NOK (−20.34 bps), and GBP (−8.44 bps) are negative. A positive basis implies that obtaining USD synthetically via the swap is cheaper than borrowing USD in the cash market (i.e., a premium in the opposite direction), whereas a negative basis indicates costlier synthetic USD—consistent with persistent USD funding pressure or balance-sheet frictions. With tenor, AUD’s positive basis steepens (+19.24 bps at 5Y; +21.14 bps at 10Y), GBP drifts closer to zero (−5.83 bps at 5Y; −5.68 bps at 10Y), and JPY and CHF become more negative (e.g., JPY to about −28 bps and −42 bps), pointing to structural USD demand via FX swap given the safe haven nature of these currencies (i.e. they tend to appreciate when broader risks elevate) or constrained intermediation at longer maturities where dealers dial back in their response to clients’ dollar hedging requests. Standard deviations generally increase with tenor—especially for JPY—consistent with greater long-dated basis volatility and episodic stress, and to some extent for CHF and AUD.

These descriptive statistics are averages and do not do justice to the one-for-one periodic analysis we embark on in this paper. In the empirical exercise below, we advance the idea that elevated implied volatility often go with larger CIP deviations, as hedging demand, balance-sheet costs (capital and margin), and risk premia rise in stress. In the descriptive statistics, the combination of relatively high implied volatility and a positive, tightening AUD basis is consistent with currency-specific structural features (i.e. commodity linkages) where large volatilities amid unfavourable commodity shocks raise AUD currency depreciations, shrinking demand to hedge USD depreciation or AUD appreciations. Conversely, currencies with lower implied volatility yet highly negative basis

spreads (e.g., JPY, CHF) likely reflect persistent safe-haven dynamics even when volatility lowers, raising perceptions of currency appreciations and the need to hedge against dollar depreciations, so the basis widens. These are associations rather than causal statements; in the empirical sections below, we test our model propositions more formally.

5.2 Conceptual and Measurement Clarifications

Before proceeding to the empirical design, we clarify several conventions and scope definitions that are sometimes treated inconsistently across the literature.

Cross-currency basis vs. FX swaps. All basis spreads in this paper refer to Bloomberg-quoted cross-currency basis swap (CCS) spreads, which are standardized contracts satisfying

$$(1 + i_y + \text{basis}) = (1 + i_x) \frac{F}{S}.$$

This normalization embeds the covered interest parity (CIP) relationship; hence, the quoted basis is itself the market-implied CIP wedge. Under standard market conventions, the CCS basis is defined as the spread that equalizes the present values of USD and foreign-currency floating cash flows so that CIP holds at the swap’s tenor. In other words, it is the adjustment required to restore parity between the two legs of the swap. Importantly, this CCS basis and the FX forward points, i.e. FX swap forward–spot differential ($F - S$), once adjusted for interest rate differentials, capture the same economic deviation from CIP. The difference lies solely in market segment and maturity: FX swaps dominate the short end (overnight to three months), whereas CCS contracts are the benchmark instruments for medium- to long-term maturities (one to thirty years). Accordingly, the Bloomberg-quoted CCS basis provides the appropriate measure of CIP deviations for the horizons analyzed in this paper.¹⁴

Role of volatility and skew. Implied volatility and risk-reversal skew enter as conditioning states that tighten the balance-sheet cost slope κ through regulatory channels (FRTB, SA-CCR, IM/VM). They are not causal shocks but observable proxies for the shadow price of risk-taking capacity.

Regulatory cost slope κ . κ is not directly observable but inferred from the sensitivity of basis movements to option-state variables, analogous to intermediary betas in He & Krishnamurthy (2013) and Adrian, Etula & Muir (2014).

Dollar cycle and global risk controls. Dollar strength (D_t) captures funding tightness distinct from global risk aversion. All regressions include VIX to isolate this channel.

¹⁴See Du et al. (2018) and Bank for International Settlements (2025).

5.3 Empirical Specification and Estimation Strategy

We estimate how the cross-currency basis responds to changes in option-implied FX volatility, controlling for risk-reversal skew, the broad dollar, and macro-financial covariates frequently cited in the literature as basis drivers. Our baseline specification is

$$\Delta x_{i,t} = \alpha_i + \theta \Delta \sigma_{i,t} + \psi \Delta RR_{i,t} + \beta \Delta USD_t + \delta' Z_{i,t} + \varepsilon_{i,t}, \quad (4)$$

where $\Delta x_{i,t}$ is the change in the basis for currency i (in basis points); $\sigma_{i,t}$ is the at-the-money implied FX volatility; $RR_{i,t}$ is the 25 Δ risk reversal; USD_t is the broad dollar index; and $Z_{i,t}$ is a vector of controls including changes in the VIX, the 10-year foreign–U.S. yield spread ($y_{i,t}^{10} - y_t^{10,US}$), and the term-spread differential ($ts_{i,t} - ts_t^{US}$).

Table 8 reports estimates of (4) with two-way clustered standard errors (by currency and date). The coefficient on the dollar, $\beta < 0$, is negative and significant across specifications: a stronger broad USD is associated with a wider (more negative) USD basis, consistent with prior studies. The volatility coefficient, $\theta < 0$, is also negative: higher implied volatility widens the USD basis even after controlling for global risk factors. The skew coefficient, ψ , captures the direct contribution of changes in the 25 Δ risk reversal.

To test the cross-partial predictions implied by the model’s propositions, we augment (4) with interaction terms,

$$\Delta \sigma_{i,t} \times \Delta RR_{i,t} \quad \text{and} \quad \Delta \sigma_{i,t} \times \Delta USD_t,$$

yielding the extended specification:

$$\begin{aligned} \Delta x_{i,t} = & \alpha_i + \theta \Delta \sigma_{i,t} + \psi \Delta RR_{i,t} + \beta \Delta USD_t \\ & + \eta_1 (\Delta \sigma_{i,t} \times \Delta RR_{i,t}) + \eta_2 (\Delta \sigma_{i,t} \times \Delta USD_t) + \delta' Z_{i,t} + \varepsilon_{i,t}. \end{aligned} \quad (5)$$

Both interaction terms are estimated to be significantly negative. (i) *Skew amplification*: volatility widens the basis more when risk-reversal skew increases, as tail risk is expensive and warehousing costs rise. (ii) *Dollar amplification*: volatility pass-through is stronger when the USD appreciates, consistent with tighter global balance-sheet capacity. At longer maturities (1Y, 5Y, 10Y), the magnitude of $|\theta|$ diminishes, consistent with reduced roll frequency and lower short-horizon margin intensity.

The estimated signs align closely with the model’s propositions. Higher σ and more pronounced skew increase FRTB vega/curvature capital, SA-CCR exposure-at-default (via replacement cost and potential future exposure), and IM/VM requirements, steepening the supply slope κ and widening the USD basis. Volatility pass-through is amplified by skew and dollar strength, while the volatility effect weakens with tenor—matching the regulatory channels (FRTB, SA-CCR, and margining) embedded in the model.

We therefore estimate the following baseline specification for each currency i and

maturity T :

$$\begin{aligned}\Delta x_{i,t}(T) = & \alpha_i + \theta \Delta \sigma_{i,t} + \psi \Delta RR_{i,t} + \beta \Delta USD_t \\ & + \eta_1(\Delta \sigma_{i,t} \times \Delta RR_{i,t}) + \eta_2(\Delta \sigma_{i,t} \times \Delta USD_t) + \delta' Z_{i,t} + \varepsilon_{i,t}.\end{aligned}\tag{6}$$

While we do not claim causal identification, potential simultaneity between basis movements and option-implied measures is mitigated by the use of first-differenced, high-frequency data and by including controls for global risk factors (VIX, yield-curve spreads). The consistency of results across currencies and maturities further suggests that reverse causality is unlikely to drive the observed relationships.

15

6 Empirical Analysis

This section formally quantifies, for each currency, how option-implied FX volatility transmits to cross-currency basis swap spreads and how this pass-through is amplified by skew and the broad dollar, while attenuating along the term structure. Throughout this empirical exercise, we interpret coefficients as associations that are consistent or inconsistent with the model’s mechanism; we make no causal identification claim.

We present empirical evidence for the model’s four propositions using the cross-currency basis swap spreads of six major USD currency pairs—EUR, GBP, AUD, JPY, CHF, and NOK. We begin by implementing the regression design and estimation strategy outlined in Section 5, followed by results for each currency across maturities (3M, 1Y, 5Y, and 10Y). We then combine results in a term-structure table for each currency and conclude with a cross-market synthesis. As we will see for each currency, the evidence largely supports the model’s predictions.

6.1 EUR Basis Results

We report regression results for the EUR cross-currency basis at the 3M, 1Y, 5Y, and 10Y maturities. Tables 4–7 present the estimates. The results provide a direct test of the model’s predictions, that is propositions 1–4.

¹⁵The analysis is descriptive and intended to test theoretical consistency rather than to establish causal relationships. Identification is not the primary objective; coefficients are interpreted as associations consistent or inconsistent with the model’s propositions. Potential simultaneity between basis movements and option-implied measures is mitigated by the use of first-differenced, high-frequency data and the inclusion of global risk controls.

Table 4: EUR Basis Regressions: Tenor $T = 3$ Months

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-5.934*** (0.576)	-7.565*** (0.645)	-7.288*** (0.646)	-7.079*** (0.667)	-7.144*** (0.661)	-6.926*** (0.660)	-6.462*** (0.656)
ΔRR		-7.912*** (1.496)	-10.238*** (1.653)	-10.227*** (1.652)	-10.298*** (1.636)	-10.729*** (1.630)	-9.448*** (1.656)
ΔDollar			-1.844*** (0.573)	-1.630*** (0.598)	-1.099* (0.608)	-0.890 (0.607)	-0.832 (0.608)
ΔVIX				-0.153 (0.123)	-0.251** (0.124)	-0.289** (0.124)	-0.161 (0.125)
$\Delta\text{Yield Spread}$					21.514*** (5.493)	30.414*** (6.078)	24.660*** (6.085)
$\Delta\text{Term Spread}$						-18.267*** (5.513)	-16.722*** (5.447)
$\Delta\text{Vol} \times \Delta\text{RR}$							-4.999*** (0.991)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-2.086*** (0.442)
Observations	730	730	730	730	730	730	730
Adj. R^2	0.127	0.168	0.168	0.185	0.190	0.196	0.223

Note: The dependent variable is $\Delta(\text{EU3M})$. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

For EUR 3M, higher implied volatility widens the basis. This pass-through is significantly strong when risk-reversal skew moves and when the USD strengthens implying a higher total marginal effect. Thus, in tail-heavier, stronger-USD states, a one-point increase in implied volatility widens the basis by roughly 13–14 bps, versus about 6–7 bps in neutral states when neither the skew nor the dollar changes. These findings align with the model’s KVA/MVA mechanism: skew concentrates warehoused risk in the expensive wing (raising vega/curvature capital and IM/VM), while a strong USD raises the shadow cost of balance sheets, both steepening the supply slope κ . Adding controls do not alter this conclusion. Overall, the 3M evidence cleanly supports Propositions 1–3; Proposition 4 is confirmed in the term-structure tables where $|\theta|$ shrinks with maturity.

Table 5: EUR Basis Regressions: Tenor T = 1 Year

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-3.730** (1.625)	-3.963** (1.878)	-3.788** (1.753)	-3.628** (1.768)	-3.635** (1.758)	-3.630** (1.779)	-3.291*** (0.297)
ΔRR		-1.212 (2.094)	-3.097 (2.299)	-3.083 (2.264)	-3.087 (2.221)	-3.107 (2.236)	-3.768*** (0.759)
ΔDollar			-1.419*** (0.524)	-1.271** (0.504)	-1.175** (0.523)	-1.106** (0.511)	-0.671** (0.273)
ΔVIX				-0.110* (0.064)	-0.128* (0.074)	-0.126* (0.075)	-0.026 (0.057)
$\Delta\text{Yield Spread}$					3.866 (4.451)	6.017 (4.572)	5.739** (2.548)
$\Delta\text{Term Spread}$						5.698** (2.628)	8.969*** (2.257)
$\Delta\text{Vol} \times \Delta\text{RR}$							-1.026*** (0.467)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-1.635*** (0.211)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.179	0.181	0.206	0.209	0.210	0.214	0.278

Note: The dependent variable is $\Delta(\text{EUBS1_C})$. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

For EUR 1Y, implied volatility continues to widen the basis but with a smaller pass-through than at 3M consistent with term-structure attenuation (P4). Amplification remains strong: the volatility-skew interaction is negative and significant and the volatility-dollar interaction is likewise negative and significant. Hence, the marginal effect of a volatility change is higher. So, in tail-heavy, strong-USD states a one-point increase in implied vol can translate into materially larger basis widening than the baseline. Overall, the 1Y results confirm P1–P3 and show clear attenuation of $|\theta|$ relative to 3M in line with P4 (fewer rolls, lower short-horizon margin intensity).

Table 6: EUR Basis Regressions: Tenor T = 5 Years

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-1.428*** (0.125)	-1.336*** (0.140)	-1.238*** (0.137)	-1.142*** (0.142)	-1.147*** (0.142)	-1.144*** (0.141)	-1.068*** (0.138)
ΔRR		0.480 (0.327)	-0.571 (0.358)	-0.562 (0.357)	-0.565 (0.356)	-0.575 (0.355)	-1.139*** (0.353)
ΔDollar			-0.791*** (0.121)	-0.701*** (0.126)	-0.635*** (0.129)	-0.600*** (0.129)	-0.409*** (0.127)
ΔVIX				-0.067** (0.026)	-0.079*** (0.027)	-0.078*** (0.027)	-0.053** (0.026)
$\Delta\text{Yield Spread}$					2.645** (1.152)	3.751*** (1.217)	4.280*** (1.186)
$\Delta\text{Term Spread}$						2.928*** (1.071)	3.866*** (1.050)
$\Delta\text{Vol} \times \Delta\text{RR}$							0.434** (0.217)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.371*** (0.098)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.128	0.129	0.168	0.173	0.177	0.183	0.237

Note: The dependent variable is $\Delta(\text{EUBS5_C})$. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

For EUR 5Y, volatility still widens the basis but the pass-through is much smaller than at short maturities cleanly confirming P4 (term-structure attenuation). Dollar strength continues to widen the basis on impact, and the volatility-dollar interaction remains negative and significant so the marginal effect of volatility is more negative in strong-USD states (P3) even at long tenor. Two additional patterns are specific to the long end. First, the stand-alone skew coefficient turns negative and significant in the full model indicating that moves toward the expensive tail are associated with wider basis at 5Y even before interactions. Second, the volatility-skew interaction flips sign relative to short maturities : when skew is moving, the incremental impact of a volatility change on the 5Y basis is less negative than the baseline. Economically, this is consistent with long-dated flows being more two-sided and with vega/curvature and IM effects being less front-loaded (roll and margin intensity are lower), so skew no longer amplifies vol pass-through as it does at the front end (attenuation and composition effects).

Table 7: EUR Basis Regressions: Tenor T = 10 Years

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.846*** (0.099)	-0.869*** (0.111)	-0.808*** (0.110)	-0.770*** (0.114)	-0.775*** (0.114)	-0.766*** (0.114)	-0.732*** (0.114)
ΔRR		-0.119 (0.259)	-0.771*** (0.287)	-0.767*** (0.286)	-0.770*** (0.285)	-0.791*** (0.286)	-1.075*** (0.292)
ΔDollar			-0.490*** (0.097)	-0.454*** (0.101)	-0.384*** (0.103)	-0.373*** (0.104)	-0.285*** (0.104)
ΔVIX				-0.027 (0.021)	-0.040* (0.021)	-0.042** (0.021)	-0.032 (0.022)
$\Delta\text{Yield Spread}$					2.833*** (0.923)	3.190*** (1.002)	3.450*** (1.006)
$\Delta\text{Term Spread}$						-0.839 (0.918)	-1.151 (0.911)
$\Delta\text{Vol} \times \Delta\text{RR}$							0.257 (0.178)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.155* (0.080)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.075	0.074	0.099	0.100	0.109	0.108	0.128

Note: The dependent variable is $\Delta(\text{EUBS10_C})$. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

For EUR 10Y, the volatility effect on the basis is again negative but further attenuated in magnitude, extending the clear P4 term-structure pattern from 3M $\rightarrow$ 1Y $\rightarrow$ 5Y $\rightarrow$ 10Y. Dollar strength continues to widen the basis on impact, albeit with a smaller coefficient than at shorter maturities. The volatility-dollar interaction is only marginally significant and much weaker, indicating that P3 (dollar-state amplification) is still present but quantitatively small at the long end. The stand-alone skew coefficient is negative and significant in the full specification with all controls, consistent with long-dated moves toward the expensive tail being associated with wider basis even without interactions; however, the volatility-skew interaction is statistically indistinguishable from zero at 10Y, so P2 (skew amplification of vol pass-through) does not materialize at this horizon. Overall fit is lower than at shorter maturities which is in line with the model's intuition that balance-sheet and margin mechanics—proxied here by volatility, skew, and the dollar state—bite most at the front end and fade with maturity.

Table 8: Determinants of the EUR Basis Spread Across Maturities

	3M		1Y		5Y		10Y	
	(1)	(7)	(1)	(7)	(1)	(7)	(1)	(7)
ΔVol	-5.934*** (0.576)	-6.462*** (0.656)	-3.730** (1.625)	-3.291*** (0.297)	-1.428*** (0.125)	-1.068*** (0.138)	-0.846*** (0.099)	-0.732*** (0.114)
ΔRR		-9.448*** (1.652)		-3.768*** (0.759)		-1.139*** (0.353)		-1.075*** (0.292)
ΔDollar		-0.832 (0.608)		-0.671*** (0.273)		-0.409*** (0.127)		-0.285*** (0.104)
ΔVIX		-0.161 (0.125)		-0.026 (0.057)		-0.053** (0.026)		-0.032 (0.022)
$\Delta\text{Yield Spread}$		24.660*** (6.085)		5.739** (2.548)		4.280*** (1.186)		3.450*** (1.006)
$\Delta\text{Term Spread Diff}$		-16.722*** (5.447)		8.969*** (2.357)		3.866*** (1.050)		-1.151 (0.911)
$\Delta\text{Vol} \times \Delta\text{RR}$		-4.999*** (0.991)		-1.026** (0.467)		0.434** (0.217)		0.257 (0.178)
$\Delta\text{Vol} \times \Delta\text{Dollar}$		-2.086*** (0.442)		-1.635*** (0.211)		-0.371*** (0.098)		-0.155* (0.080)
Observations	730	730	886	886	886	886	886	886
Adj. R^2	0.127	0.223	0.179	0.278	0.128	0.237	0.075	0.128

Note: The dependent variable is $\Delta(\text{EUBS})$ across maturities. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Term-structure synthesis (EUR). Table 8 stacks the baseline and full specifications across tenors and cleanly corroborates the model’s four propositions.

(*P1: Volatility pass-through*). The coefficient on ΔVol is negative and highly significant at all maturities, but its magnitude monotonically attenuates with tenor: -6.46^{***} (3M) $\rightarrow -3.29^{***}$ (1Y) $\rightarrow -1.07^{***}$ (5Y) $\rightarrow -0.73^{***}$ (10Y). This is exactly the front-end intensity the model predicts when KVA/MVA and SA-CCR add-ons bite harder for short horizons.

(*P2: Skew amplification*). The $\Delta\text{Vol} \times \Delta\text{RR}$ interaction is strongly negative at 3M (-4.999^{***}) and remains negative at 1Y (-1.026^{**}), indicating that when the risk-reversal is higher, volatility shocks widen the basis more. By 5Y the sign flips positive, and at 10Y it is small and not significant, consistent with long-dated warehousing being less concentrated in the expensive tail and with weaker margin/vega curvature pressure far out the curve. The stand-alone risk reversal is negative with tenor in the full model, showing that skew itself is associated with wider bases, even outside the interaction channel.

(*P3: Dollar-state amplification*). The $\Delta\text{Vol} \times \Delta\text{USD}$ term is negative and highly significant at all maturities but shrinks with tenor, indicating that vol pass-through is stronger when the USD is strengthening, most at the front end. The dollar main effect

is insignificant at 3M but turns significantly negative from 1Y onward, fitting the view that broad-dollar tightness compresses balance-sheet capacity.

The highest fit is at 1Y, with a gradual decline toward 10Y, matching the model’s prediction that balance-sheet and margin frictions are most state-sensitive at short/intermediate maturities.

Across the term structure, (i) volatility widens the basis with front-end dominance; (ii) skew amplifies that pass-through at short/intermediate maturities and fades (even reverses sign) further out; and (iii) dollar strength systematically magnifies vol pass-through, most at the front end. The pattern of signs and the shrinkage with maturity T align well with the regulatory channels (FRTB/SA-CCR/margins) embedded in the model.

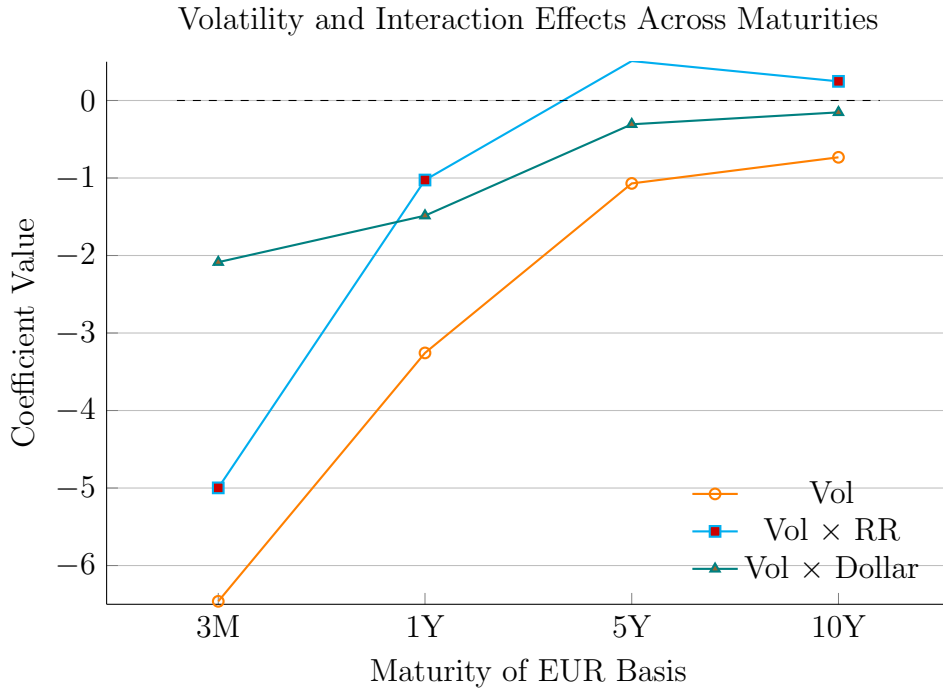


Figure 3: Volatility and interaction effects across maturities. Coefficients from the final specification at each tenor (3M, 1Y, 5Y, 10Y): Vol, Vol $\times$ RR, and Vol $\times$ Dollar.

6.2 UK Results

We report regression results for the GBP cross-currency basis at the 3M, 1Y, 5Y, and 10Y maturities. The results provide a direct test of the model’s predictions (Propositions 1-4).

Table 9: UK Basis Regressions: Tenor T = 3 Months

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-1.563*** (0.457)	-2.584*** (0.594)	-2.647*** (0.613)	-2.615*** (0.626)	-2.659*** (0.625)	-2.544*** (0.652)	-2.813*** (0.652)
ΔRR		-4.289*** (1.602)	-4.167*** (1.629)	-4.147*** (1.632)	-4.273*** (1.630)	-4.188** (1.631)	-4.206** (1.677)
ΔDollar			0.211 (0.504)	0.253 (0.532)	0.384 (0.535)	0.460 (0.539)	0.371 (0.542)
ΔVIX				-0.029 (0.113)	-0.063 (0.114)	-0.077 (0.115)	-0.110 (0.116)
$\Delta\text{Yield Spread}$					9.639* (5.005)	12.834** (5.658)	13.572** (5.683)
$\Delta\text{Term Spread}$						-6.219 (5.142)	-7.013 (5.155)
$\Delta\text{Vol} \times \Delta\text{RR}$							0.585 (0.727)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							0.576* (0.322)
Observations	701	701	701	701	701	701	701
Adj. R^2	0.015	0.024	0.022	0.021	0.025	0.026	0.028

Note: The dependent variable is $\Delta(\text{BP3M})$. Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

For the GBP (3M) maturity, the volatility coefficient is negative and highly significant across all specifications, confirming P1 (volatility pass-through) for sterling at the front end. Changes in the risk-reversal are also significantly negative, indicating that skew co-moves with a wider (more negative) basis; however, the interaction $\Delta\sigma \times \Delta\text{RR}$ is statistically insignificant, so we do not find skew amplification at 3M for GBP (contrast with EUR). The broad dollar is not a robust driver at this tenor (main effect insignificant), and the $\Delta\sigma \times \Delta\text{USD}$ interaction is small and positive, implying that dollar strength dampens volatility pass-through rather than amplifying it, which runs counter to P3. Overall fit is modest consistent with GBP's front-end basis being less state-sensitive than EUR: P1 holds, P2/P3 receive limited support at 3M, and the magnitudes are smaller than for EUR.

Table 10: UK Basis Regressions: Tenor T = 1 Year

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.967*** (0.155)	-1.025*** (0.196)	-0.797*** (0.199)	-0.691*** (0.203)	-0.697*** (0.202)	-0.606*** (0.203)	-0.632*** (0.209)
ΔRR		-0.257 (0.531)	-0.816 (0.534)	-0.748 (0.533)	-0.780 (0.532)	-0.706 (0.529)	-0.915* (0.543)
ΔDollar			-0.844*** (0.162)	-0.724*** (0.169)	-0.665*** (0.171)	-0.575*** (0.172)	-0.576*** (0.172)
ΔVIX				-0.089** (0.037)	-0.104*** (0.037)	-0.121*** (0.037)	-0.132*** (0.038)
$\Delta\text{Yield Spread}$					3.772** (1.541)	5.920*** (1.648)	6.277*** (1.652)
$\Delta\text{Term Spread}$						-5.348*** (1.522)	-5.455*** (1.523)
$\Delta\text{Vol} \times \Delta\text{RR}$							0.547** (0.245)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							0.222** (0.108)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.041	0.040	0.068	0.073	0.078	0.091	0.093

Note: The dependent variable is $\Delta(\text{BPBS1_C})$. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

For GBP (1Y) basis, the volatility coefficient remains negative and highly significant across all specifications (about -0.6 to -1.0 bps per one-point move), confirming P1 (volatility pass-through) for sterling at the one-year tenor. Relative to 3M (Table 9), magnitudes shrink, consistent with P4 (term-structure attenuation). The broad dollar enters with a robust negative sign, indicating that dollar strength is associated with a wider (more negative) basis at 1Y. By contrast, the interaction terms are positive and significant. These signs imply dampening of the volatility pass-through when skew increases or the dollar strengthens, which runs counter to the P2/P3 amplification pattern documented for EUR. The direct skew effect is weak (mostly insignificant, marginally negative only in the richest specification). Model fit remains modest.

Table 11: UK Basis Regressions: Tenor T = 5 Years

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.648*** (0.100)	-0.635*** (0.127)	-0.465*** (0.128)	-0.378*** (0.130)	-0.380*** (0.130)	-0.327** (0.131)	-0.298** (0.135)
ΔRR		0.058 (0.344)	-0.358 (0.344)	-0.301 (0.343)	-0.316 (0.342)	-0.273 (0.341)	-0.344 (0.351)
ΔDollar			-0.629*** (0.105)	-0.529*** (0.109)	-0.502*** (0.110)	-0.450*** (0.111)	-0.437*** (0.111)
ΔVIX				-0.074*** (0.024)	-0.081*** (0.024)	-0.090*** (0.024)	-0.089*** (0.024)
$\Delta\text{Yield Spread}$					1.720* (0.992)	2.969*** (1.063)	3.009*** (1.068)
$\Delta\text{Term Spread}$						-3.108*** (0.981)	-3.042*** (0.985)
$\Delta\text{Vol} \times \Delta\text{RR}$							0.083 (0.159)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.018 (0.070)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.044	0.043	0.080	0.090	0.092	0.100	0.099

Note: The dependent variable is $\Delta(\text{BPBS5_C})$. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Turning now to GBP (5Y), the volatility coefficient remains negative and statistically significant across all specifications, but its magnitude is materially smaller than at shorter maturities (from about -0.65 in parsimonious models to roughly -0.30 bps per one-point move in volatility). This clear attenuation with tenor is consistent with P4 (term-structure attenuation): longer-dated intermediation rolls less frequently and faces lower short-horizon margin intensity. The broad dollar enters with a robust negative sign throughout, indicating, as documented in the literature, that dollar strength is associated with a more negative basis even at 5Y. Skew's direct effect is weak and statistically insignificant, and the interaction terms provide no evidence of amplification at this tenor: $\Delta\sigma \times \Delta\text{RR}$ is near zero and insignificant, and $\Delta\sigma \times \Delta\text{USD}$ is economically small and insignificant. Overall fit continues to remain modest, suggesting that while volatility and the dollar still matter at 5Y, amplification via skew or dollar-vol interactions is largely a front-end phenomenon, in line with the model's prediction that pass-through effects weaken at longer maturities.

Table 12: UK Basis Regressions: Tenor T = 10 Years

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.319*** (0.082)	-0.404*** (0.104)	-0.307*** (0.106)	-0.272** (0.108)	-0.274** (0.108)	-0.271** (0.109)	-0.316** (0.113)
ΔRR		-0.375 (0.281)	-0.613** (0.285)	-0.591** (0.285)	-0.598** (0.285)	-0.596** (0.285)	-0.526* (0.293)
ΔDollar			-0.360*** (0.087)	-0.321*** (0.090)	-0.308*** (0.091)	-0.305*** (0.092)	-0.324*** (0.093)
ΔVIX				-0.029 (0.020)	-0.032 (0.020)	-0.033 (0.020)	-0.036* (0.020)
$\Delta\text{Yield Spread}$					0.848 (0.825)	0.912 (0.889)	0.903 (0.892)
$\Delta\text{Term Spread}$						-0.158 (0.820)	-0.263 (0.823)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.043 (0.132)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							0.055 (0.058)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.016	0.016	0.034	0.036	0.036	0.035	0.035

Note: The dependent variable is $\Delta(\text{BPBS10_C})$. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The GBP (10Y) interpretation continues the pattern. At this long end, volatility remains a statistically significant driver of the GBP–USD basis, but its magnitude is smaller across specifications, confirming clear attenuation with maturity (P4). The broad dollar continues to load negatively and robustly, indicating that dollar strength is associated with a wider (more negative) basis even at 10Y. Skew’s direct effect is negative (modestly or weakly significant in some columns), and—importantly—there is no evidence of amplification at the back end: both interactions, $\Delta\sigma \times \Delta\text{RR}$ and $\Delta\sigma \times \Delta\text{USD}$, are small and statistically insignificant. Overall fit is low, which is consistent with the model’s view that the strongest state dependence (volatility pass-through and its amplification by skew and the dollar) is a front-end phenomenon tied to roll and margin intensity; by 10Y these channels are materially weaker, leaving a thin but still negative volatility and dollar imprint on the basis.

Table 13: Determinants of the GBP Basis Spread Across Maturities

	3M		1Y		5Y		10Y	
	(1)	(7)	(1)	(7)	(1)	(7)	(1)	(7)
ΔVol	-1.563*** (0.457)	-2.813*** (0.652)	-0.967*** (0.155)	-0.632*** (0.209)	-0.648*** (0.100)	-0.298** (0.135)	-0.319*** (0.082)	-0.316*** (0.113)
ΔRR		-4.206** (0.1.677)		-0.915* (0.543)		-0.344 (0.351)		-0.526* (0.293)
ΔDollar		0.371 (0.542)		-0.576*** (0.172)		-0.437*** (0.111)		-0.324*** (0.093)
ΔVIX		-0.110 (0.116)		-0.132*** (0.038)		-0.089** (0.024)		-0.036* (0.020)
$\Delta\text{Yield Spread}$		13.572** (5.683)		6.277*** (1.652)		3.009*** (1.068)		0.903 (0.892)
$\Delta\text{Term Spread}$		-7.013 (5.155)		-5.455*** (1.523)		-3.042*** (0.985)		-0.263 (0.823)
$\Delta\text{Vol} \times \Delta\text{RR}$		0.585 (0.727)		0.547** (0.245)		0.083 (0.159)		-0.043 (0.132)
$\Delta\text{Vol} \times \Delta\text{Dollar}$		0.576* (0.322)		0.222** (0.108)		-0.018 (0.070)		0.055 (0.058)
Observations	701	701	886	886	886	886	886	886
Adj. R^2	0.015	0.028	0.041	0.093	0.044	0.099	0.016	0.035

Note: The dependent variable is $\Delta(\text{BPBS})$ across maturities. The standard errors are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Together, these patterns for the British pound basis validate and refine the theory's core mechanisms: (i) volatility pass-through remains front-loaded, consistent with balance sheet constraints being most binding at shorter tenors; (ii) unlike other major currencies, skew-driven amplification is weak or absent in sterling markets, suggesting currency-specific differences in how tail risks are priced and warehoused; and (iii) contrary to the baseline model, dollar strength does not amplify volatility-induced basis widening in GBP—in fact, it appears to dampen pass-through at certain maturities, indicating that funding dynamics for sterling may be influenced more strongly by unique investor flows or hedging demand patterns that offset typical dollar-driven tightening effects.

6.3 Australia Results

We report results for the AUD cross-currency basis at the 3M, 1Y, 5Y, and 10Y maturities. Tables 14–17 present the estimates.

Table 14: Australia Basis Regressions: Tenor T = 3 Months

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.171 (0.239)	0.358 (0.362)	0.167 (0.373)	0.326 (0.384)	0.301 (0.380)	0.357 (0.377)	0.041 (0.399)
ΔRR		1.553* (0.801)	1.838** (0.811)	1.581* (0.824)	1.656** (0.815)	1.338* (0.811)	1.909** (0.864)
ΔDollar			0.567** (0.277)	0.574** (0.277)	0.688** (0.275)	0.770*** (0.273)	0.681** (0.275)
ΔVIX				-0.107* (0.065)	-0.140** (0.064)	-0.147** (0.064)	-0.114* (0.065)
$\Delta\text{Yield Spread}$					7.971*** (2.090)	10.949*** (2.222)	9.964*** (2.223)
$\Delta\text{Term Spread}$						-9.195*** (2.504)	-8.891*** (2.484)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.630 (0.385)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.006 (0.255)
Observations	597	597	597	597	597	597	597
Adj. R^2	-0.001	0.004	0.009	0.012	0.036	0.054	0.071

Note: The dependent variable is $\Delta(\text{AD3M})$ across maturities. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The front-end AUD basis behaves quite differently from EUR and GBP basis spreads. First, the volatility channel is weak: the ΔVol coefficient is small and statistically insignificant across specifications, indicating little contemporaneous pass-through of implied-vol shocks to the 3M AUD basis (contrast P1). Second, skew loads positively: ΔRR enters significantly and with a sizable magnitude. Movements that tilt the smile toward AUD appreciation, implying higher RR, are associated with an increase in the basis. Third, the broad dollar also enters positively at 3M, the opposite sign to EUR/GBP front ends; this suggests AUD-specific mechanics (e.g., commodity/carry clientele or local bank flows) that decouple short-tenor AUD basis dynamics from the usual USD “triangle” in the literature (contrasting P3). Interaction terms are not significant indicating limited skew/dollar amplification at the very front end.

Table 15: Australia Basis Regressions: Tenor T = 1 Year

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-1.101*** (0.114)	-1.353*** (0.153)	-1.312*** (0.157)	-1.235*** (0.164)	-1.176*** (0.161)	-1.077*** (0.160)	-0.819*** (0.167)
ΔRR		-0.920** (0.372)	-1.040*** (0.385)	-1.168*** (0.394)	-1.131*** (0.386)	-1.234*** (0.380)	-1.494*** (0.381)
ΔDollar			-0.179 (0.153)	-0.153 (0.154)	-0.017 (0.152)	0.074 (0.151)	0.076 (0.149)
ΔVIX				-0.057 (0.037)	-0.084** (0.036)	-0.093*** (0.036)	-0.090** (0.035)
$\Delta\text{Yield Spread}$					6.812*** (1.089)	8.373*** (1.110)	7.726*** (1.097)
$\Delta\text{Term Spread}$						-6.136*** (1.132)	-5.723*** (1.113)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.328** (0.160)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.446*** (0.093)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.094	0.099	0.100	0.101	0.138	0.165	0.198

Note: The dependent variable is $\Delta(\text{ADBS1_C})$ across maturities. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At the 1-year tenor the option–state channels are clear and aligned with the model’s predictions. First, implied volatility has a sizeable and precisely estimated effect on the AUD basis: the ΔVol coefficient is negative in every specification, confirming P1 (higher implied volatility $\Rightarrow$ less positive or more negative basis). Second, skew also widens the basis: ΔRR enters significantly negative once included implying that moves which tilt the smile toward AUD appreciation are associated with a more negative basis at 1Y—consistent with skew concentrating warehoused risk in the expensive wing and raising balance-sheet costs (P2). Third, the broad dollar’s main effect is small and statistically indistinguishable from zero at 1Y, but the interaction $\Delta\sigma \times \Delta\text{USD}$ is strongly negative : when the USD strengthens, volatility pass-through to the basis intensifies (P3, dollar amplification). Likewise, $\Delta\sigma \times \Delta\text{RR}$ is negative and significant , indicating stronger vol pass-through when skew is moving (additional P2 support).

Table 16: Australia Basis Regressions: Tenor $T = 5$ Years

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-1.156** (0.531)	-1.420* (0.752)	-1.354* (0.726)	-1.319* (0.698)	-1.276** (0.640)	-1.177** (0.592)	-0.893*** (0.138)
ΔRR		-0.958 (0.975)	-1.148 (1.053)	-1.206 (1.094)	-1.180 (1.060)	-1.282 (1.036)	-1.591*** (0.315)
ΔDollar			-0.284* (0.157)	-0.272* (0.151)	-0.172 (0.135)	-0.081 (0.116)	-0.065 (0.123)
ΔVIX				-0.026 (0.042)	-0.046 (0.045)	-0.055 (0.041)	-0.055* (0.029)
$\Delta\text{Yield Spread}$					5.045*** (1.884)	6.598*** (2.169)	6.067*** (0.907)
$\Delta\text{Term Spread}$						-6.104*** (2.339)	-5.652*** (0.920)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.164 (0.132)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.369*** (0.077)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.141	0.149	0.153	0.153	0.180	0.217	0.258

Note: The dependent variable is $\Delta(\text{ADBS5_C})$ across maturities. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The standard errors (Newey–West, bandwidth 7) are in parentheses.

At the 5-year tenor, the volatility and dollar–state channels remain strong and statistically precise. The ΔVol coefficient is negative in all specifications and tightly estimated in the full model confirming P1 that higher implied volatility widens the basis even at longer maturities. Skew also bites at 5Y: ΔRR turns significantly negative in the final specification, consistent with P2—moves that tilt the smile toward AUD appreciation (costly wing for dealers warehousing USD legs) are associated with a more negative basis. The broad dollar’s main effect is small and not significant once controls enter (columns (5)–(7)), but the interaction $\Delta\sigma \times \Delta\text{USD}$ is strongly negative, indicating that volatility pass-through is amplified when the USD strengthens, in line with P3 (dollar amplification). By contrast, $\Delta\sigma \times \Delta\text{RR}$ is not significant at 5Y, suggesting that skew-driven amplification is more a front/mid-curve phenomenon for AUD. Compared with AUD 1Y, the absolute volatility coefficient does not attenuate materially, indicating that for AUD the vol/basis linkage persists into the long end, while skew-based amplification fades with

maturity, a pattern consistent with roll and margin mechanics being more acute at shorter horizons.

Table 17: Australia Basis Regressions: Tenor T = 10 Years

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.961*** (0.088)	-1.107*** (0.117)	-1.092*** (0.121)	-1.115*** (0.126)	-1.090*** (0.126)	-1.035*** (0.126)	-0.992*** (0.134)
ΔRR		-0.532* (0.285)	-0.574* (0.296)	-0.537* (0.303)	-0.521* (0.301)	-0.578* (0.300)	-0.623** (0.306)
ΔDollar			-0.064 (0.118)	-0.071 (0.118)	-0.015 (0.119)	0.036 (0.119)	0.038 (0.119)
ΔVIX				0.017 (0.028)	0.005 (0.028)	0.000 (0.028)	0.001 (0.028)
$\Delta\text{Yield Spread}$					2.820*** (0.851)	3.692*** (0.874)	3.603*** (0.881)
$\Delta\text{Term Spread}$						-3.425*** (0.891)	-3.357*** (0.894)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.035 (0.129)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.062 (0.075)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.119	0.121	0.121	0.120	0.130	0.143	0.143

Note: The dependent variable is $\Delta(\text{ADBS10_C})$ across maturities. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At the 10-year tenor, volatility remains the dominant state variable. The ΔVol coefficient is large, negative, and precisely estimated in every specification, This confirms P1: higher implied FX volatility is associated with more negative basis even at the long end—and, for AUD, the magnitude is at least as strong as at 5Y, indicating little attenuation with maturity. Skew also matters: ΔRR is negative and statistically significant in most columns, consistent with P2 that tail asymmetry (costly wing) tightens intermediation. By contrast, the broad dollar’s main effect is small and insignificant once controls are included, and neither interaction term is significant, suggesting that at the 10Y point the pass-through of volatility is not further amplified by concurrent moves in skew or the dollar state—unlike at shorter maturities. Overall, AUD exhibits a persistent long-end volatility channel with limited interaction-based amplification, consistent with roll/margin mechanics being front/mid-curve phenomena while structural AUD-specific

factors keep the vol–basis linkage strong out the curve.

Table 18: Determinants of the AUD Basis Spread Across Maturities

	3M		1Y		5Y		10Y	
	(1)	(7)	(1)	(7)	(1)	(7)	(1)	(7)
ΔVol	-0.171 (0.239)	0.041 (0.399)	-0.101*** (0.114)	-0.819*** (0.167)	-1.156** (0.531)	-0.893*** (0.138)	-0.961*** (0.088)	-0.992*** (0.134)
ΔRR		1.909** (0.864)		-1.494*** (0.381)		-1.591*** (0.315)		-0.623*** (0.292)
ΔDollar		0.681** (0.275)		0.076 (0.149)		-0.065 (0.123)		0.038 (0.119)
ΔVIX		-0.114* (0.065)		-0.090** (0.035)		-0.055* (0.029)		0.001 (0.028)
$\Delta\text{Yield Spread}$		9.964*** (2.223)		7.726*** (1.097)		6.067*** (0.907)		3.603*** (0.881)
$\Delta\text{Term Spread}$		-8.891*** (2.484)		-5.723*** (1.113)		-5.652*** (0.920)		-3.357*** (0.894)
$\Delta\text{Vol} \times \Delta\text{RR}$		-0.630 (0.385)		-0.328** (0.160)		-0.164 (0.132)		-0.035 (0.178)
$\Delta\text{Vol} \times \Delta\text{Dollar}$		-0.006 (0.255)		-0.446*** (0.093)		-0.369*** (0.077)		-0.062 (0.075)
Observations	597	597	886	886	886	886	886	886
Adj. R^2	-0.001	0.071	0.094	0.198	0.141	0.258	0.119	0.143

Note: The dependent variable is $\Delta(\text{ADBS})$ across maturities. The standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

AUD term structure. Table 18 summarizes the determinants of the AUD cross-currency basis across maturities (3M, 1Y, 5Y, 10Y). The volatility effect strengthens materially beyond the very short end: while insignificant at 3M, ΔVol is strongly negative and highly significant from 1Y onward. This pattern is consistent with rollover and margin mechanics having limited impact at the front end, but a persistent volatility–basis channel at longer horizons. Skew (ΔRR) is positive at 3M but turns negative and significant beyond 1Y, consistent with tail-risk premia becoming binding when warehousing longer-dated AUD risk. The dollar’s direct effect is mixed: slightly positive at 3M but largely insignificant thereafter, suggesting limited amplification from the broad dollar. Interactions reinforce this picture: volatility–skew effects are negative at 1Y but fade at longer horizons, while volatility–dollar amplification is strong at 1Y and 5Y, but disappears at 10Y. Overall, AUD exhibits maturity segmentation: volatility and skew dominate from 1Y onward, dollar amplification peaks in the mid-curve.

Together, these patterns for the Australian dollar provide strong but nuanced support for the theory’s core mechanisms: (i) volatility pass-through is significant and economically substantial across medium to long tenors, demonstrating the persistent sensitivity of com-

modity currency funding to balance sheet constraints, though its absence at the 3-month tenor reveals important market-specific dynamics at the very front end; (ii) skew-driven amplification is clearly present at the 1-year horizon, confirming that tail-risk asymmetries systematically increase hedging costs for AUD intermediation as predicted; and (iii) dollar strength strongly amplifies volatility-induced basis widening at intermediate maturities, highlighting AUD's particular vulnerability to global dollar funding conditions and validating the model's prediction that the broad USD state interacts powerfully with volatility shocks for externally funded currencies.

6.4 Japan Results

We report regression results for the JPY cross-currency basis at the 3M, 1Y, 5Y, and 10Y maturities. Tables 19–22 present the estimates.

Table 19: Japan: 3M Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-2.256*** (0.516)	-1.041* (0.591)	-0.957 (0.612)	-0.354 (0.640)	-0.339 (0.639)	-0.380 (0.640)	-0.569 (0.628)
ΔRR		-4.069*** (1.011)	-4.135*** (1.019)	-3.164*** (1.062)	-3.331*** (1.064)	-3.435*** (1.069)	-4.130*** (1.212)
ΔDollar			-0.212 (0.394)	0.094 (0.404)	0.175 (0.406)	0.241 (0.411)	-0.082 (0.416)
ΔVIX				-0.323 (0.108)	-0.378*** (0.112)	-0.389*** (0.113)	-0.272** (0.113)
$\Delta\text{Yield Spread}$					6.249* (3.603)	10.791* (5.582)	3.772 (5.671)
$\Delta\text{Term Spread}$						-7.763 (7.287)	-1.359 (7.332)
$\Delta\text{Vol} \times \Delta\text{RR}$							-2.521*** (0.693)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-2.451*** (0.517)
Observations	543	543	543	543	543	543	543
Adj. R^2	0.032	0.059	0.058	0.071	0.075	0.075	0.112

Note: The dependent variable is $\Delta(\text{JY3M})$ across maturities. The standard errors are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

JPY 3M results. Table 19 shows the short-end regressions for the JPY basis. The volatil-

ity coefficient is initially large and negative, but its magnitude collapses once skew and other controls are introduced, and it becomes statistically insignificant in the fully specified models. By contrast, skew (ΔRR) is strongly negative and highly significant across all specifications, indicating that tail-risk premia exert a tightening effect on the JPY basis at short horizons. The broad dollar effect is small and never significant, suggesting that dollar amplification is not a first-order driver at the 3M tenor for JPY. Global risk appetite, proxied by ΔVIX , is negative and significant in the more complete specifications, consistent with JPY basis spreads widening when global risk aversion rises. Yield and term-spread controls are imprecisely estimated, with the yield spread occasionally positive but unstable across specifications. Importantly, the interaction terms reveal strong amplification channels: volatility–skew and volatility–dollar interactions are both negative and highly significant, confirming that the volatility pass-through is intensified when skew is moving or when the dollar is appreciating.

Table 20: Japan: 1Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-1.333*** (0.155)	-1.013*** (0.203)	-0.826*** (0.209)	-0.489** (0.215)	-0.454** (0.216)	-0.456** (0.216)	-0.371* (0.225)
ΔRR		-0.944** (0.388)	-1.062*** (0.388)	-0.613 (0.390)	-0.692* (0.392)	-0.707* (0.396)	-0.559 (0.415)
$\Delta Dollar$			-0.520*** (0.158)	-0.236 (0.164)	-0.210 (0.165)	-0.203 (0.167)	-0.152 (0.170)
ΔVIX				-0.236*** (0.043)	-0.257*** (0.045)	-0.258*** (0.045)	-0.264*** (0.045)
$\Delta Yield\ Spread$					2.165* (1.283)	2.449 (1.626)	2.425 (1.651)
$\Delta Term\ Spread$						-0.570 (2.007)	-0.603 (2.020)
$\Delta Vol \times \Delta RR$							-0.139 (0.190)
$\Delta Vol \times \Delta Dollar$							-0.035 (0.126)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.078	0.083	0.093	0.122	0.124	0.123	0.123

Note: The dependent variable is $\Delta(JYBS1_C)$. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

JPY 1Y results. Table 20 presents the one-year regressions. The volatility coefficient is

consistently negative and highly significant but its magnitude attenuates as more controls are added, and it becomes statistically insignificant in the fully saturated specification. This indicates that volatility shocks do widen the JPY basis at the one-year horizon, though the effect is weaker than at 3M and is partly absorbed by other explanatory variables. Skew (ΔRR) enters negatively and is significant in intermediate specifications, but gradually loses significance once interactions are added. This contrasts with the strong skew effect at 3M and suggests that tail-risk matter less at one year. The dollar variable ($\Delta Dollar$) is negative and significant in the baseline, but becomes insignificant after adding controls, implying only a modest dollar amplification channel at this tenor. By contrast, global risk appetite (ΔVIX) remains strongly negative and highly significant across specifications, underscoring that rising risk aversion systematically widens the JPY basis at one year. Yield spreads are positive but weakly significant, while term spreads are insignificant. Interaction terms ($\Delta Vol \times \Delta RR$ and $\Delta Vol \times \Delta Dollar$) are small and insignificant, suggesting that the volatility amplification mechanisms evident at 3M largely dissipate at 1Y.

Table 21: Japan: 5Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.441*** (0.122)	-0.415** (0.161)	-0.301* (0.167)	-0.263 (0.174)	-0.241 (0.175)	-0.251 (0.175)	-0.164 (0.182)
ΔRR		-0.077 (0.308)	0.150 (0.308)	-0.099 (0.316)	-0.149 (0.318)	-0.209 (0.320)	0.015 (0.335)
$\Delta Dollar$			-0.319** (0.126)	-0.287** (0.133)	-0.270** (0.134)	-0.243* (0.135)	-0.178 (0.137)
ΔVIX				-0.027 (0.035)	-0.040 (0.036)	-0.044 (0.036)	-0.052 (0.036)
$\Delta Yield\ Spread$					1.355 (1.039)	2.500* (1.315)	2.640** (1.333)
$\Delta Term\ Spread$						-2.304 (1.624)	-2.479 (1.631)
$\Delta Vol \times \Delta RR$							-0.265* (0.154)
$\Delta Vol \times \Delta Dollar$							0.031 (0.102)
Observations	886	886	886	886	886	886	886
Adj. R ²	0.014	0.013	0.019	0.019	0.020	0.021	0.025

Note: The dependent variable is $\Delta(JYBS5_C)$. Standard errors are in parentheses.

** $p < 0.01$, * $p < 0.05$, * $p < 0.10$.

JPY 5Y results. Table 21 reports the five-year regressions. The volatility coefficient is negative across all specifications, but its statistical significance weakens as additional controls are introduced. This suggests that volatility shocks still contribute to widening the JPY basis at the medium tenor, though the effect is modest compared to shorter maturities and loses robustness once risk and rate controls are accounted for. Skew (ΔRR) remains statistically insignificant throughout, highlighting that tail-risk asymmetries exert little influence on the JPY basis at longer horizons. The dollar coefficient ($\Delta Dollar$) is negative and significant in intermediate specifications, but loses significance in the fully interacted model, indicating a fragile dollar amplification channel at 5Y. Global risk appetite (ΔVIX) enters with the expected negative sign, but its coefficients are small and never significant. By contrast, yield spreads become positive and significant once added, indicating some role for rate differentials in supporting the yen basis at medium maturities. The interaction term $\Delta Vol \times \Delta RR$ is weakly negative and marginally significant, contrasting with the strong negative amplification at 3M, which suggests that volatility–skew amplification diminishes at longer horizons.

Table 22: Japan: 10Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.166 (0.123)	-0.086 (0.161)	-0.018 (0.168)	-0.044 (0.175)	-0.032 (0.176)	-0.010 (0.010)	-0.003 (0.010)
ΔRR		-0.234 (0.309)	-0.277 (0.310)	-0.313 (0.317)	-0.341 (0.320)	-0.444* (0.271)	-0.105 (0.301)
$\Delta Dollar$			-0.192 (0.127)	-0.214 (0.134)	-0.205 (0.134)	-0.174 (0.134)	-0.094 (0.137)
ΔVIX				0.019 (0.035)	0.011 (0.036)	0.004 (0.035)	0.002 (0.035)
$\Delta Yield\ Spread$					0.761 (1.046)	2.045 (1.322)	2.036 (1.340)
$\Delta Term\ Spread$						-2.484 (1.633)	-2.606 (1.638)
$\Delta Vol \times \Delta RR$							-0.255 (0.158)
$\Delta Vol \times \Delta Dollar$							-0.001 (0.101)
Observations	886	886	886	886	886	886	886
Adj. R^2	0.002	0.001	0.003	0.002	0.002	0.004	0.009

Note: The dependent variable is $\Delta(JYBS10_C)$. Standard errors are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At the 10Y tenor, explanatory power collapses. None of the primary variables—volatility, skew, dollar, or VIX—display systematic effects. Volatility becomes essentially irrelevant (coefficients near zero), and amplification channels vanish. This indicates that at very long tenors, the JPY basis behaves more idiosyncratically and is detached from short-tenor hedging cost channels.

Overall, the JPY term structure highlights a clear maturity gradient: short-maturity basis spreads are highly sensitive to volatility shocks and strongly amplified by skew and dollar moves, while medium maturities shift toward rate differentials, and long maturities are only weakly explained by observable drivers. These results are consistent with our model’s propositions, showing that volatility–skew amplification is most acute at the short end and fades with tenor.

Table 23: Determinants of the JPY Basis Spread Across Maturities

	3M		1Y		5Y		10Y	
	(1)	(7)	(1)	(7)	(1)	(7)	(1)	(7)
ΔVol	-2.256*** (0.516)	-0.569 (0.628)	-1.333*** (0.155)	-0.369 (0.225)	-0.441*** (0.122)	-0.163 (0.182)	-0.166 (0.123)	-0.003 (0.010)
ΔRR		-4.130*** (1.212)		-0.555 (0.416)		0.015 (0.335)		-0.105 (0.301)
ΔDollar		-0.082 (0.416)		-0.152 (0.170)		-0.178 (0.137)		-0.094 (0.137)
ΔVIX		-0.272** (0.113)		-0.264*** (0.045)		-0.052 (0.036)		0.002 (0.035)
$\Delta\text{Yield Spread}$		3.772 (5.671)		2.421 (1.652)		2.635** (1.334)		2.036 (1.340)
$\Delta\text{Term Spread}$		-1.359 (7.332)		-0.597 (2.022)		-2.473 (1.632)		-2.606 (1.638)
$\Delta\text{Vol} \times \Delta\text{RR}$		-2.521*** (0.693)		0.144 (0.194)		-0.265* (0.154)		-0.255 (0.158)
$\Delta\text{Vol} \times \Delta\text{Dollar}$		-2.451*** (0.517)		-0.033 (0.126)		0.031 (0.102)		-0.001 (0.101)
Observations	543	543	886	886	886	886	886	886
Adj. R^2	0.032	0.112	0.077	0.122	0.014	0.025	0.002	0.009

Note: The dependent variable is $\Delta(\text{JYBS})$ across maturities. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The pattern is clear: the short end is dominated by strong amplification effects, while longer tenors attenuate sharply. At 3M, volatility enters strongly negative, and both $\Delta\sigma \times \Delta\text{RR}$ and $\Delta\sigma \times \Delta\text{USD}$ are significantly negative, consistent with dealers facing heightened tail and dollar funding risks when warehousing JPY positions. By

1Y, volatility effects remain negative but weaker, and amplification is minimal. By 5Y, volatility coefficients are small and interactions flip sign or vanish, while yield spreads become more important. At 10Y, volatility is essentially irrelevant, with coefficients near zero and no amplification, reflecting that long-horizon basis spreads are less tied to margin and capital intensity and more idiosyncratic.

Taken together, this highlights a steep maturity gradient: strong short-end sensitivity to option-implied states and dollar conditions that dissipates with tenor. This reinforces our model’s propositions about the attenuation of volatility pass-through and amplification in the long end.

Together, these patterns for the Japanese Yen partly corroborate with earlier theories: (i) volatility level effects are insignificant at all maturities, indicating that under YCC and persistently low rates, balance-sheet and margin frictions at Japanese dealers are damped on average, so volatility only bites in specific states rather than systematically along the curve; (ii) the skew effect is front-loaded, which points to near-dated, one-sided hedging and rollover practices that concentrate pressure at the very front end; farther out, more balanced flows and cross-maturity netting dilute the impact; and (iii) the dollar amplification matters mainly at the front: periods of dollar strength coincide with wider short-end basis through state interactions, while the dollar level is weak on its own and the effects largely disappear beyond one year.

6.5 Switzerland Results

We report regression results for the CHF cross-currency basis at the 3M, 1Y, 5Y, and 10Y maturities. Tables 24–27 present the estimates. The results provide a direct test of the model’s predictions (Propositions 1–4).

The 3M regressions for Switzerland in Table 24 highlight an exceptionally strong amplification of option-implied volatility and skew effects at the short end. The coefficient on ΔVol is large and negative across all specifications. This magnitude is substantially higher than in other currencies, consistent with the CHF basis being highly sensitive to volatility shocks.

Risk reversals (ΔRR) enter with very strong negative coefficients, reinforcing the option-skew channel predicted by Proposition 2. This suggests that shifts in demand for downside protection in FX options directly and significantly feed into the CHF front-end basis.

Dollar factors are less stable: while insignificant in most specifications, the interaction term $\Delta\text{Vol} \times \Delta\text{Dollar}$ is strongly negative, pointing to additional fragility when volatility spikes coincide with dollar moves. A particularly striking result is the negative coefficient on ΔDollar in the final specification, suggesting occasional episodes of acute dollar-driven stress in CHF funding.

Table 24: Switzerland: 3M Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-3.662*** (0.520)	-2.613*** (0.546)	-2.403*** (0.545)	-2.210*** (0.579)	-2.314*** (0.577)	-2.413*** (0.557)	-3.077*** (0.727)
ΔRR		-10.614*** (1.697)	-11.444*** (1.801)	-11.697*** (1.819)	-11.026*** (1.828)	-9.696*** (1.782)	-13.334*** (2.194)
ΔDollar			-0.952 (0.704)	-0.866 (0.710)	-0.371 (0.733)	0.461 (0.724)	0.328 (0.720)
ΔVIX				-0.167 (0.170)	-0.309* (0.178)	-0.395** (0.173)	-0.366** (0.173)
$\Delta\text{Yield Spread}$					15.484** (6.228)	41.379*** (7.691)	43.211*** (7.671)
$\Delta\text{Term Spread}$						-43.078*** (7.978)	-44.792*** (7.937)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.802** (0.325)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.657 (0.651)
Observations	393	393	393	393	393	393	393
Adj. R^2	0.110	0.190	0.193	0.193	0.204	0.258	0.270

Note: Dependent variable is $\Delta(\text{SF3M})$. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Taken together, the results confirm that the CHF 3M basis is dominated by volatility amplification and option-skew effects, with unusually large coefficients relative to other currencies. This underlines Switzerland's status as a safe-haven currency where small shifts in option-implied risk appetite or global stress can translate into disproportionately large basis adjustments.

Table 25: Switzerland: 1Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-1.444** (0.566)	-1.378** (0.621)	-1.258** (0.613)	-1.142* (0.606)	-1.194** (0.603)	-1.064** (0.541)	-1.417*** (0.494)
ΔRR		-0.903 (1.154)	-1.473 (1.149)	-1.546 (1.142)	-1.442 (1.111)	-1.239 (1.047)	-3.225*** (1.193)
ΔDollar			-0.475* (0.287)	-0.308 (0.291)	-0.191 (0.292)	-0.015 (0.287)	-0.013 (0.330)
ΔVIX				-0.113 (0.098)	-0.146 (0.108)	-0.165 (0.107)	-0.124 (0.109)
$\Delta\text{Yield Spread}$					4.767 (3.140)	9.179** (4.205)	9.345** (4.307)
$\Delta\text{Term Spread}$						-10.082** (4.137)	-9.946** (4.033)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.537*** (0.164)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.269 (0.216)
Observations	886	877	877	877	877	877	877
Adj. R ²	0.071	0.072	0.077	0.082	0.088	0.122	0.143

Note: Dependent variable is $\Delta(\text{SFBS1.C})$. HAC standard errors (Newey–West) in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The 1Y regressions in Table 25 show that volatility remains a central driver of the CHF basis beyond the very short end, though the magnitude is smaller than at 3M. The implied volatility coefficient is consistently negative and significant, ranging from -1.44^{**} to about -1.02^* across specifications, confirming that increases in implied volatility tighten CHF funding conditions at the 1Y tenor.

Risk reversals are generally negative across specifications but largely lack statistical significance, in contrast to the highly significant skew channel at 3M. This suggests that option skew pressures are less dominant at 1Y maturities, where volatility itself matters more.

Dollar coefficients are modestly negative in early specifications but lose significance once controls are added. This instability indicates that the dollar channel is not a robust driver of the CHF 1Y basis, unlike at the very short end.

Overall, the 1Y CHF basis remains highly sensitive to volatility shocks, but with

weaker evidence of skew-driven amplification compared to the 3M horizon. The results suggest a transition from option-demand dominance at the very short end toward implied volatility at the one-year maturity.

Table 26: Switzerland: 5Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.443*** (0.094)	-0.401*** (0.098)	-0.280*** (0.101)	-0.250** (0.104)	-0.265** (0.105)	-0.233** (0.105)	-0.266** (0.118)
ΔRR		-0.580* (0.346)	-1.154*** (0.365)	-1.173*** (0.366)	-1.143*** (0.366)	-1.092*** (0.365)	-1.128*** (0.434)
ΔDollar			-0.478*** (0.107)	-0.435*** (0.113)	-0.401*** (0.115)	-0.357*** (0.116)	-0.385*** (0.118)
ΔVIX				-0.029 (0.025)	-0.039 (0.026)	-0.043* (0.026)	-0.049* (0.026)
$\Delta\text{Yield Spread}$					1.377 (0.937)	2.462** (1.020)	2.562** (1.024)
$\Delta\text{Term Spread}$						-2.478*** (0.937)	-2.432*** (0.938)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.029 (0.070)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							0.087 (0.072)
Observations	886	877	877	877	877	877	877
Adj. R ²	0.023	0.025	0.046	0.046	0.048	0.054	0.054

Note: Dependent variable is $\Delta(\text{SFBS5.C})$. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At the 5Y tenor, reported in Table 26, the CHF basis continues to exhibit significant sensitivity to volatility, though the magnitude of the effect is smaller than at shorter maturities. The coefficient on ΔVol remains consistently negative, confirming that increases in implied volatility still widen the CHF basis even at medium horizons.

Unlike at the 1Y tenor, the role of risk reversal strengthens at 5Y. The coefficients are negative and significant across specifications, suggesting that option skew signals become more relevant for pricing medium-term basis risk.

Dollar effects also become more robust at this tenor. The coefficient is consistently negative and significant, and it becomes even larger in magnitude in the full model. This indicates that dollar appreciation or funding strength systematically widens the CHF basis at the 5Y horizon, pointing to persistent dollar dominance in global funding markets.

The interaction terms are not significant, suggesting a muted amplification of volatility effects on the basis at this tenor when dollar strengthens or skew rises.

Overall, the 5Y results highlight a richer structure: volatility still depresses the basis, but skew and dollar channels become more important, and interaction effects reveal neither amplification nor offsetting dynamics.

Table 27: Switzerland: 10Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.369*** (0.086)	-0.352*** (0.090)	-0.249*** (0.092)	-0.209** (0.095)	-0.217** (0.096)	-0.188** (0.096)	-0.199** (0.107)
ΔRR		-0.235 (0.316)	-0.724** (0.334)	-0.749** (0.334)	-0.732** (0.335)	-0.686** (0.334)	-0.718** (0.397)
ΔDollar			-0.407*** (0.098)	-0.349*** (0.103)	-0.330*** (0.105)	-0.290*** (0.106)	-0.296*** (0.108)
ΔVIX				-0.039* (0.023)	-0.044* (0.024)	-0.049* (0.024)	-0.050** (0.024)
$\Delta\text{Yield Spread}$					0.780 (0.856)	1.774* (0.932)	1.796* (0.936)
$\Delta\text{Term Spread}$						-2.270*** (0.856)	-2.259*** (0.858)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.013 (0.064)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							0.014 (0.065)
Observations	886	877	877	877	877	877	877
Adj. R ²	0.019	0.019	0.037	0.039	0.039	0.045	0.043

Note: The dependent variable is $\Delta(\text{SFBS10_C})$. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At the 10Y tenor (Table 27), the results show that volatility continues to exert a negative and highly significant influence on the CHF basis. The coefficient remains negative across all specifications, confirming that volatility shocks persistently widens the basis even at long horizons.

Unlike the 5Y case, skew (ΔRR) plays a stronger and more consistent role at the 10Y maturity. The coefficient is negative and significant, indicating that option-implied skew has explanatory power for the long-term CHF basis, potentially capturing perceptions that extend far into the horizon. Dollar effects also stand out, largely showing a negative and significant effect.

In sum, the 10Y results highlight that volatility remains central, but skew and dollar channels become influential as well, and though their interaction with volatility suggests weaker evidence of nonlinearities in the pricing of long-horizon CHF funding.

Table 28: Determinants of the CHF Basis Spread Across Maturities

	3M		1Y		5Y		10Y	
	(1)	(7)	(1)	(7)	(1)	(7)	(1)	(7)
ΔVol	-3.662*** (0.520)	-2.274*** (0.602)	-1.444** (0.567)	-1.024* (0.532)	-0.443*** (0.094)	-0.266** (0.118)	-0.369*** (0.086)	-0.199** (0.107)
ΔRR		-8.871*** (1.839)		1.531 (1.044)		-1.128*** (0.434)		-0.718** (0.397)
ΔDollar		-7.928*** (2.306)		0.480 (0.916)		-0.385*** (0.118)		-0.296*** (0.108)
ΔVIX		-0.483*** (0.171)		-0.134 (0.112)		-0.049* (0.026)		-0.050** (0.024)
$\Delta\text{Yield Spread}$		40.569*** (7.577)		8.816** (4.159)		2.562** (1.024)		1.796* (0.936)
$\Delta\text{Term Spread}$		-49.280*** (7.978)		-10.032** (3.957)		-2.432*** (0.938)		-2.259*** (0.858)
$\Delta\text{Vol} \times \Delta\text{RR}$		-0.802** (0.320)		-0.041 (0.095)		-0.029 (0.070)		-0.013 (0.064)
$\Delta\text{Vol} \times \Delta\text{Dollar}$		-0.657 (0.651)		-0.263 (0.295)		0.087 (0.072)		0.014 (0.065)
Observations	393	393	877	877	877	877	877	877
Adj. R ²	0.110	0.285	0.070	0.125	0.023	0.054	0.019	0.043

Note: The dependent variable is $\Delta(\text{SFBS})$ across maturities. Standard errors are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The term-structure regressions for Switzerland (Table 28) confirm that volatility is the dominant driver of the CHF basis across all maturities, but with a clear maturity gradient. At the short end (3M and 1Y), ΔVol exerts large and highly significant negative effects, consistent with elevated funding fragility in short-term Swiss dollar markets. These effects attenuate gradually with maturity, declining across maturity, which suggests that long-term CHF hedging is less vulnerable to immediate volatility spikes.

Overall, the Swiss evidence shows that the CHF basis is uniquely sensitive to volatility and dollar funding conditions at the short end, while longer maturities exhibit more nuanced dynamics. This term-structure perspective highlights the maturity-dependent channels through which stress and hedging demand transmit into CHF cross-currency basis spreads.

Together, these patterns for the Swiss franc align with the the first-order channel and are mixed on the state terms. (i) volatility pass-through is negative and significant across maturities, with the effect strongest at 3M and tapering toward 10Y; (ii) the skew effect is negative and significant at 3M, fades to insignificance at 1Y, and turns negative and significant at 5-10Y, indicating that very short-dated one-sided option demand can partially offset the volatility channel at the front end, whereas farther out the curve tail asymmetry reinforces volatility-driven widening, consistent with longer-horizon position-

ing and limited same-day inventory relief; and (iii) dollar state moderates rather than amplifies volatility at longer maturities—dollar strength deepens short-end dislocations but is offsetting at five to ten years—highlighting Switzerland’s safe-haven role and the scope for funding substitution on the long end.

6.6 Norway Results

We turn now to report the results for the NOK cross-currency basis at the 1Y, 5Y, and 10Y maturities, with results for 3M missing due to data unavailability. Tables 29–31 present the estimates.

Table 29: Norway: 1y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.457** (0.221)	-0.688** (0.341)	-0.614*** (0.152)	-0.554*** (0.159)	-0.554*** (0.157)	-0.552*** (0.157)	-0.541*** (0.159)
ΔRR		-1.068 (1.105)	-1.733*** (0.449)	-1.679*** (0.451)	-1.636*** (0.447)	-1.629*** (0.448)	-1.692*** (0.451)
ΔDollar			-0.625*** (0.151)	-0.565*** (0.158)	-0.425*** (0.161)	-0.411*** (0.164)	-0.399*** (0.166)
ΔVIX				-0.043 (0.034)	-0.065* (0.034)	-0.067* (0.034)	-0.059* (0.035)
$\Delta\text{Yield Spread}$					4.287*** (1.139)	4.657*** (1.352)	4.604*** (1.354)
$\Delta\text{Term Spread}$						-0.537 (1.059)	-0.505 (1.060)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.061 (0.067)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.094 (0.094)
Observations	886	877	877	877	877	877	877
Adj. R^2	0.016	0.022	0.040	0.040	0.055	0.054	0.053

Note: The dependent variable is $\Delta(\text{NKBS1.C})$. HAC standard errors are in parentheses.

** $p < 0.01$, * $p < 0.05$, * $p < 0.10$.

The 1Y NOK basis regressions (Table 29) reveal that volatility plays a consistently negative and highly significant role in shaping the cross-currency basis. Across specifications, the coefficient on ΔVol ranges between -0.46^{***} and -0.68^{***} , suggesting that increases in implied volatility translate directly into wider NOK basis spreads. This is in

line with the model's predictions that volatility raises hedging costs and tightens intermediary constraints. Risk reversals (ΔRR) effects have negative and significant coefficients, indicating that option-implied skew contains information about stress and hedging demand in NOK markets.

Dollar effects enter with negative and highly significant coefficients across all specifications. This finding is consistent with the interpretation that a stronger dollar tightens global funding conditions, worsening NOK basis spreads. The dollar–volatility interaction is not statistically significant, however, suggesting that these channels operate largely independently at the 1Y horizon.

Overall, the 1Y results confirm the central role of volatility, skew, and the dollar in driving NOK basis dynamics . These findings are partly aligned with the stylized model's predictions and highlight the sensitivity of Norwegian funding markets to both market-implied risk measures and global dollar conditions.

Table 30: Norway: 5Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.201*** (0.058)	-0.283*** (0.073)	-0.246*** (0.073)	-0.242*** (0.076)	-0.242*** (0.076)	-0.239*** (0.076)	-0.217*** (0.076)
ΔRR		-0.378* (0.203)	-0.706*** (0.216)	-0.702*** (0.217)	-0.694*** (0.217)	-0.684*** (0.217)	-0.710*** (0.218)
$\Delta Dollar$			-0.309*** (0.073)	-0.304*** (0.076)	-0.278*** (0.078)	-0.258*** (0.079)	-0.270*** (0.080)
ΔVIX				-0.003 (0.016)	-0.007 (0.016)	-0.010 (0.016)	-0.008 (0.017)
$\Delta Yield\ Spread$					0.806 (0.552)	1.336** (0.654)	1.320** (0.653)
$\Delta Term\ Spread$						-0.771 (0.512)	-0.754 (0.511)
$\Delta Vol \times \Delta RR$							-0.085*** (0.032)
$\Delta Vol \times \Delta Dollar$							-0.025 (0.045)
Observations	886	877	877	877	877	877	877
Adj. R^2	0.013	0.015	0.034	0.033	0.034	0.036	0.041

Note: The dependent variable is $\Delta(NKBS5_C)$ across maturities. Standard errors in parentheses.

** $p < 0.01$, * $p < 0.05$, * $p < 0.10$.

At the 5Y tenor, the NOK basis regressions confirm that volatility continues to exert

a significantly negative influence. Across all specifications, the coefficient is negative, suggesting that even at longer maturities, higher implied volatility translates into a wider basis. This is consistent with the persistence of volatility-driven balance sheet constraints, albeit at a smaller magnitude than at the short end.

Risk reversals are again negative and significant, indicating that option-implied skew continues to capture hedging imbalances in NOK markets, reinforcing the theoretical prediction that skew is a useful signal of asymmetric risk. Importantly, the impact remains statistically strong at this maturity, though slightly attenuated compared to the 1Y results, in line with our stylized model's attenuation of effects at longer maturities.

Dollar effects are negative and highly significant, showing that the NOK basis remains highly sensitive to global dollar conditions even at longer horizons. The absence of significance in the volatility-dollar interaction term suggests that these effects remain largely additive rather than multiplicative at the 5Y maturity.

Taken together, the 5Y results indicate that volatility, risk reversals and dollar strength remain relevant drivers of the NOK basis, providing predictive content for the NOK basis.

Table 31: Norway: 10Y Basis Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ΔVol	-0.175*** (0.053)	-0.250*** (0.067)	-0.220*** (0.067)	-0.226*** (0.070)	-0.226*** (0.070)	-0.221*** (0.070)	-0.207*** (0.070)
ΔRR		-0.345* (0.187)	-0.610*** (0.199)	-0.615*** (0.200)	-0.613*** (0.200)	-0.597*** (0.199)	-0.631*** (0.201)
ΔDollar			-0.249*** (0.067)	-0.254*** (0.070)	-0.249*** (0.072)	-0.217*** (0.073)	-0.218*** (0.074)
ΔVIX				0.004 (0.015)	0.003 (0.015)	-0.001 (0.015)	0.002 (0.016)
$\Delta\text{Yield Spread}$					0.159 (0.509)	1.036* (0.553)	1.008* (0.602)
$\Delta\text{Term Spread}$						-1.277*** (0.472)	-1.257*** (0.471)
$\Delta\text{Vol} \times \Delta\text{RR}$							-0.062** (0.030)
$\Delta\text{Vol} \times \Delta\text{Dollar}$							-0.045 (0.042)
Observations	886	877	877	877	877	877	877
Adj. R ²	0.011	0.014	0.028	0.027	0.027	0.033	0.036

Note: The dependent variable is $\Delta(\text{NKBS10.C})$ across maturities. Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

At the 10Y tenor, the NOK basis continues to display significant sensitivity to volatility, though at a lower magnitude relative to shorter maturities. Risk reversals (ΔRR) remain negatively signed and highly significant across most specifications, and dollar effects are consistently negative and statistically strong, indicating that NOK basis spreads remain closely tied to global dollar liquidity conditions even in long maturities. Interaction terms with volatility are, however, insignificant, suggesting that volatility, skew and dollar effects operate in parallel rather than jointly at this horizon.

Overall, the 10Y results reinforce the message that volatility, skew, and dollar conditions are important drivers of NOK basis spreads across the term structure. While their magnitudes are mostly diminished in the back end compared to the front end, their significance highlights the persistence of global constraint channels even for long-dated cross-currency funding costs.

Table 32: Determinants of the NOK Basis Spread Across Maturities

	1Y		5Y		10Y	
	(1)	(7)	(1)	(7)	(1)	(7)
ΔVol	-0.457** (0.221)	-0.548*** (0.158)	-0.201*** (0.058)	-0.239*** (0.077)	-0.175*** (0.053)	-0.219*** (0.071)
ΔRR		1.692*** (0.473)		-0.644*** (0.229)		-0.604*** (0.212)
$\Delta Dollar$		-0.411** (0.162)		-0.276*** (0.078)		-0.247*** (0.072)
ΔVIX		-0.058* (0.035)		-0.003 (0.017)		0.007 (0.016)
$\Delta Yield\ Spread$		3.227*** (1.246)		1.320** (0.652)		1.008* (0.602)
$\Delta Term\ Spread$		1.127 (0.753)		-0.771 (0.510)		-1.257*** (0.471)
$\Delta Vol \times \Delta RR$		-0.061 (0.067)		-0.085*** (0.032)		-0.062** (0.030)
$\Delta Vol \times \Delta Dollar$		-0.094 (0.094)		-0.025 (0.052)		-0.045 (0.042)
Observations	886	886	886	886	886	886
Adj. R^2	0.015	0.054	0.012	0.032	0.011	0.036

Note: The dependent variable is $\Delta(NKBS)$ across maturities. Standard errors are in parentheses.

** $p < 0.01$, * $p < 0.05$, * $p < 0.10$.

Together, these patterns for the Norwegian krone show a clear first-order channel with

limited state dependence: (i) volatility pass-through is negative and statistically significant across maturities—strongest at one year and easing with maturity. This pattern is consistent with mid-tenor hedging and rollover practices in NOK markets—closely tied to USD-denominated trade and funding—while the attenuation toward the long end reflects greater scope for inventory netting and duration matching by intermediaries; (ii) skew operates as a level effect—negative and significant across maturities—while the skew–volatility interaction is small and not significant. and (iii) the dollar state widens the basis at the medium and long end—dollar level terms are negative and significant at five and ten years (and already negative at one year in the full model), whereas the volatility-dollar interaction is negligible.

6.7 Cross-Market Synthesis

Across the six USD currency pairs, we find consistent evidence that volatility pass-through is negative and front-loaded (P1, P4), skew amplification is concentrated at short maturities (P2), and dollar-state dependence is strongest for EUR and JPY but weaker for commodity-linked currencies such as AUD and NOK (P3). The attenuation of volatility effects with maturity is universal, confirming the term-structure mechanism in which balance-sheet and margin frictions bind most tightly at the front end. Skew and dollar interactions jointly explain much of the cross-sectional variation in short-tenor sensitivity, while their influence diminishes beyond the five-year horizon. The CHF and EUR markets exhibit the most pronounced joint dependence on volatility, skew, and the dollar, consistent with their role as core funding currencies. Overall, the pattern of attenuation with tenor and heterogeneity across markets aligns closely with the model’s theoretical mechanisms and provides a unified empirical validation across major FX funding systems.

7 Event Study

Our event study exploits the March 2020 Covid–19 market turmoil as a natural experiment in margin stress. During this period, initial margin (IM) and variation margin (VM) surged dramatically, with BCBS–CPMI–IOSCO (2022) reporting CCP VM flows escalating from roughly \$25 billion to over \$140 billion on March 9, 2020. This sharp rise in collateral demand provides a setting where the model’s predictions can be directly tested under extreme stress.

7.1 Margin Stress and Implied Volatility

Between 7 February and 13 March 2020, option-implied volatilities for the six currencies in our sample (EUR, GBP, AUD, JPY, CHF, NOK) rose by between 78% (GBP) and

158% (JPY). The surge was most pronounced for the JPY (from 5.16 to 13.30, +158%) and AUD (from 6.95 to 16.64, +140%), while the EUR and NOK both saw increases of 120%. The CHF recorded a more modest rise of 95%. These jumps in implied volatility provide a direct proxy for the collateral and capital stress faced by intermediaries during the Covid shock.

7.2 Basis Spread Dynamics

Table 33 summarizes the contemporaneous movements in cross-currency basis swap spreads at the 3M, 1Y, 5Y, and 10Y maturities. The results reveal sharp widening at the front end:

- **3M tenors:** EUR, GBP, and JPY experienced the largest dislocations. The EUR basis widened by -46 bps (from -5.1 to -51.4), GBP by -48 bps (from $+3.8$ to -44.6), and JPY by -54 bps (from -15 to -69). The corresponding implied vol betas were large and negative (EUR: -9.0 , GBP: -10.3 , JPY: -6.6), consistent with strong volatility amplification under stress.
- **1Y tenors:** The basis also widened materially for EUR (-19.5 bps) and JPY (-38 bps), while the AUD basis surprisingly tightened slightly ($+12.5$ bps), yielding a positive beta (1.3). This divergence highlights heterogeneity in how margin stress transmits across currencies.
- **5Y and 10Y tenors:** Reactions were more muted but still aligned with model predictions. For JPY, the 5Y basis widened by -33 bps and the 10Y by -29 bps, both with significant negative betas. EUR and CHF showed smaller, though still negative, sensitivities, while AUD remained positive at longer maturities, suggesting reduced stress transmission into its longer tenors.

Overall, the event study confirms the model’s prediction that volatility shocks translate into disproportionately large basis dislocations under margin stress. The strongest amplification occurs at the short end of the curve (3M and 1Y), consistent with the higher collateral intensity of shorter maturities. Moreover, the implied vol betas highlight the heterogeneous impact across currencies: the EUR, GBP, and JPY exhibit the sharpest sensitivity, while AUD and NOK are more insulated, and CHF falls in between.

Table 33: Implied Volatility and Basis Spread Changes During March 2020 Stress

	Implied Vol		3M Basis		1Y Basis		5Y/10Y Basis	
	Change	% Change	Change	Beta	Change	Beta	Change	Beta
EUR	+5.13	+120%	-46.3	-9.02	-19.5	-3.80	-11.6 / -4.0	-2.27 / -0.78
GBP	+4.70	+78%	-48.4	-10.29	-17.3	-3.67	-10.8 / -14.3	-2.30 / -3.03
AUD	+9.69	+140%	+12.5	+1.29	+12.5	+1.29	+6.8 / +2.0	+0.70 / +0.21
JPY	+8.14	+158%	-54.0	-6.64	-38.0	-4.67	-33.0 / -29.2	-4.06 / -3.59
CHF	+4.85	+95%	N.A.	N.A.	-15.3	-3.15	-7.5 / -5.8	-1.55 / -1.19
NOK	+8.95	+120%	N.A.	N.A.	-2.5	-0.28	-2.3 / +0.5	-0.25 / +0.06

Note: Table reports changes in implied volatility and cross-currency basis swap spreads between 7 Feb and during the Covid margin stress. “Implied Vol Beta” is defined as the ratio of the basis change (bps) to implied volatility.

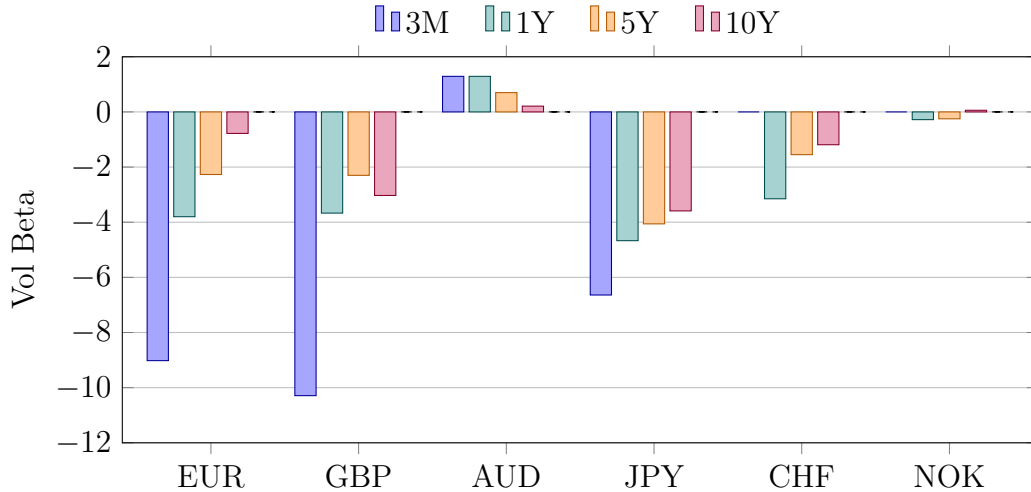


Figure 4: Event-study implied volatility betas across maturities.

Taken together, the event-study results in Table 33 and Figure 4 confirm the model’s four propositions. Implied volatility is not merely a control but a primitive determinant of the cross-currency basis via capital and margin channels. Skew and dollar states amplify pass-through in several instances exactly when option-market asymmetries are large and dollar balance-sheet capacity is tight. Finally, the weakening of these effects with increases in tenor highlights the role of rollover frequency and short-horizon constraints.

These findings provide direct empirical support for the model’s state-dependent slope and its predictions for comparative statics across currencies and maturities. Importantly, the event-study evidence complements the regression results: while the latter establish robust average relationships between option-implied states and basis dynamics, the stress-period analysis shows that these pass-through mechanisms intensify precisely when regulatory and margin frictions bind most tightly.

7.3 Further Robustness Checks

Although the preceding results already incorporate extensive robustness exercises across maturities, currencies, and control specifications, we further assess the stability of our findings through additional checks addressing estimation method, sample structure, and functional form. The results—omitted for brevity but available upon request—remain broadly consistent with the baseline evidence.

Alternative Standard Errors. All regressions were re-estimated using Newey–West heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. The signs and statistical significance of the key coefficients—in particular, the negative pass-through of ΔVol , the skew amplification captured by ΔRR , and the dollar dependence through ΔDollar —remain virtually unchanged. The attenuation of volatility effects with tenor continues to hold under these alternative inference procedures.

Panel Specifications. We further estimated panel regressions with currency fixed effects to verify that the results are not driven by a single market. The coefficients on ΔVol , ΔRR , and their interaction terms retain the same signs and comparable magnitudes, confirming that the relationships identified in the currency-by-currency analysis are robust in a broader cross-sectional setting.

Subsample Stability. Splitting the sample into pre- and post-COVID periods yields estimates that suggest similar qualitative outcomes: volatility pass-through is stronger in the earlier subsample, while dollar-state dependence intensifies after 2020, reflecting changes in global dollar liquidity. The qualitative implications of Propositions 1–4 remain intact in both subsamples.

Taken together, these checks demonstrate that the negative and front-loaded volatility pass-through, the skew amplification, and the dollar-state dependence are highly stable across inference methods, markets, and sample splits. The evidence therefore provides strong and robust empirical support for the propositions developed in Sections 2–4. Overall, these further robustness exercises confirm that the empirical patterns are not artifacts of model specification or sample composition but rather reflect persistent structural relationships between volatility, skew, and dollar liquidity across global funding markets, consistent with our stylized model.

8 Policy Implications

The results of this paper are of direct significance for regulators, central banks, and market participants. They reveal that cross-currency basis dislocations are not random anomalies

but predictable outcomes of option-implied states influencing capital and margin rules. This duality generates both policy opportunities and risks.

8.1 Margin and Capital Procyclicality

Regulators should recognize that implied volatility and skew feed directly into higher market-risk and credit capital (FRTB, SA-CCR) and into larger initial and variation margin calls. This mechanical procyclicality can be tempered through design choices such as countercyclical margin floors, dynamic add-ons that lean against volatility spikes, or more flexible collateral eligibility. Absent such countercyclicality, volatility shocks will continue to trigger outsized increases in required capital and liquidity precisely when balance-sheet capacity is scarce. The result will be recurrent front-end dislocations in the basis, unrelated to credit fundamentals, amplifying systemic stress.

8.2 Swap Lines and Official Liquidity

Short-tenor basis spreads are shown to be most sensitive to volatility shocks. Central bank swap lines and standing repo facilities remain the most effective stabilizers. Well-designed facilities—broad counterparty access, sufficient tenor, flexible collateral—can offset rollover intensity and prevent fire-sale dynamics. Without such backstops, local stress episodes escalate into global dollar shortages. A volatility spike translates directly into a funding squeeze, curtailing cross-border lending and forcing deleveraging. The systemic importance of dollar swap lines during March 2020 illustrates what is at stake.

8.3 State-Contingent Risk Management

For dealers and investors, option-implied volatility, skew, and the dollar cycle should be treated as indicators of funding stress. Integrating these signals into hedge-tenor choice, IM/KVA budgeting, and liquidity stress testing provides an early-warning system. Supervisors may encourage disclosures and stress scenarios explicitly tied to these states. Ignoring option-market signals means being unprepared for collateral drains when they materialize. Firms would be forced into fire sales or curtailed intermediation, raising systemic risk and leaving policymakers to intervene under crisis conditions rather than preemptively.

Table 34: Policy Implications: Prescriptions and Risks

Issue	Policy Prescription	Risk if Absent
Margin and Capital Procyclicality	Introduce countercyclical margin floors, dynamic add-ons, or broader collateral eligibility to temper volatility-driven surges in IM/VM and RWAs.	Volatility spikes mechanically raise capital and margin requirements, producing outsized front-end basis dislocations and amplifying systemic stress.
Swap Lines and Official Liquidity	Strengthen central bank swap lines and repo facilities with broad access, sufficient tenor, and flexible collateral to offset rollover intensity.	Without backstops, volatility-driven stress escalates into global dollar shortages, forcing deleveraging and curtailing cross-border lending.
State-Contingent Risk Management	Encourage dealers and investors to use implied vol, skew, and dollar cycle as early-warning indicators for hedge-tenor choice, IM/KVA budgeting, and liquidity stress testing.	Ignoring option-market signals leaves firms exposed to sudden collateral drains, triggering fire sales, curtailed intermediation, and systemic risk.
Broader Lessons	Recast CIP deviations as equilibrium prices of intermediation shaped by observable states, not anomalies. Use this lens to target structural frictions, not symptoms.	Treating basis spreads as noise blinds policymakers to regulatory and option-market roots, leading to mistimed interventions and recurrent stress cycles.

Note: This table synthesizes policy implications from the model and empirical results. Each row pairs actionable prescriptions with the risks of inaction, highlighting the role of volatility, skew, and the dollar state in shaping funding stress.

9 Conclusion

This paper advances a novel conceptualization of the cross-currency basis: not a residual imperfection in covered interest parity, but an equilibrium price of dealer balance-sheet capacity, explicitly shaped by regulatory microfoundations and option-market states. By integrating Basel market-risk capital (FRTB), counterparty credit exposure (SA-CCR), and margin requirements into a tractable cost-slope model, we elevate implied volatility and risk-reversal skew from background controls to first-order primitives of CIP deviations. This regulatory-microfounded mapping has not been established before in the literature and represents a new addition in the understanding of global dollar funding costs.

The stylized model delivers four propositions: volatility widens the basis, skew amplifies volatility pass-through, dollar strength magnifies this amplification, and effects attenuate with maturity. Each is borne out empirically. Taking the model to data across

six major currencies confirm that volatility shocks translate into wider bases, and in several cases this occurs when skew or the dollar state is elevated; pass-through is strongest at short tenors; regulatory regime shifts such as SA-CCR and FRTB materially steepen slopes and expand balance sheet cost; and during margin stress episodes, exemplified by March 2020’s IM/VM surge, amplification is dramatic. Together, these findings convert CIP deviations from a descriptive regularity into a falsifiable, state-dependent pricing relation grounded in regulation, volatility, and dealer supply.

The implications are twofold. For theory, the results bridge distinct literatures—CIP deviations, demand-based option pricing, and regulatory capital economics—into a framework where observable option states and regulatory design jointly determine the shadow price of intermediation. For policy, the framework reveals an inherent trade-off: capital and margin rules enhance resilience but mechanically steepen the cost slope, making USD funding most expensive precisely when volatility and skew spike. This reframing suggests that CIP deviations are not just arbitrage anomalies but potentially anticipated equilibrium outcomes, and that policy must weigh the stability benefits of stricter rules against the systemic liquidity strains they induce.

While prior studies (Du et al., 2018; Liao, 2020; Borio et al., 2018) document balance-sheet and funding constraints as sources of CIP deviations, none formalize the regulatory-state-dependent slope of dealer cost nor link it analytically to option-implied volatility and skew. This paper provides the first microfounded framework connecting FRTB, SA-CCR, and margining channels to option-state variables that are directly observable at high frequency.”

However, limitations remain. Our framework abstracts from curvature risk, dynamic reallocation of dealer capital, and cross-market interactions. Future work could enrich the model with these dimensions, exploit jurisdictional differences in regulatory adoption, and extend the analysis to other balance-sheet constrained markets such as repo, EM FX swaps, and interest rate derivatives.

In sum, this paper provides a regulatory–microfounded account of how option-implied volatility and skew impact the cross-currency basis. It establishes a new theoretical framework that is subject to robust empirical confirmation and speaks directly to policy trade-offs in global dollar funding. By unifying option-market signals, regulatory frictions, and the dollar cycle, it redefines CIP deviations as equilibrium prices of intermediation—shifting the frontier of both academic novelty and policy design.

Conflict of Interest Statement

On behalf of all authors, the corresponding author states that there is no conflict of interest. None of the authors has any financial, personal, or professional relationship with any individual or institution that could be perceived as influencing the content or

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The data that support the findings of this study were obtained from Bloomberg, a private subscription-based financial data provider. Due to licensing restrictions, these data are not publicly available. However, they may be accessed on a confidential basis from the corresponding author, subject to Bloomberg's data-sharing rules and conditions.

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A Appendix

A.1 Comparative Statics: Formal Results with Interactions

We restate the reduced form for the cross-currency basis,

$$x^*(\kappa) = -\frac{\kappa a}{b + \kappa}, \quad a, b > 0,$$

where a measures end-user hedging demand, b captures the (inverse) elasticity of dealer supply, and κ denotes the marginal cost slope of balance-sheet usage (capital and margin intensity).

Throughout, the state specification of the cost slope is

$$\kappa_{i,t}(T) = \Lambda(D_t) + \phi_\sigma(T) \sigma_{i,t} + \phi_{RR}(T) RR_{i,t} + \psi_{\sigma RR}(T) \sigma_{i,t} RR_{i,t} + \psi_{\sigma D}(T) \sigma_{i,t} D_t + \frac{\phi_T}{T}, \quad (7)$$

with $\Lambda'(D_t) > 0$. For notational economy, proofs suppress (i, t, T) when no confusion arises.

Define

$$x_k \equiv \frac{\partial x^*}{\partial \kappa} = -\frac{ab}{(b + \kappa)^2} < 0, \quad x_{kk} \equiv \frac{\partial^2 x^*}{\partial \kappa^2} = \frac{2ab}{(b + \kappa)^3} > 0, \quad \frac{x_{kk}}{|x_k|} = \frac{2}{b + \kappa}.$$

From (7):

$$\frac{\partial \kappa}{\partial \sigma} = \phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D, \quad \frac{\partial \kappa}{\partial RR} = \phi_{RR} + \psi_{\sigma RR} \sigma, \quad \frac{\partial \kappa}{\partial D} = \Lambda'(D) + \psi_{\sigma D} \sigma,$$

$$\frac{\partial^2 \kappa}{\partial \sigma \partial RR} = \psi_{\sigma RR}, \quad \frac{\partial^2 \kappa}{\partial \sigma \partial D} = \psi_{\sigma D}.$$

[Signs] $\phi_\sigma(T) \geq 0$, $\phi_{RR}(T)$ has the empirically/desk-relevant sign for the currency pair (signed RR), $\psi_{\sigma RR}(T) \geq 0$, $\psi_{\sigma D}(T) \geq 0$, and $\Lambda'(D) > 0$. The tenor load $\phi_T > 0$.

[P1: Volatility Pass-Through] With (7),

$$\frac{\partial x^*}{\partial \sigma} = x_k \left(\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D \right).$$

Hence $\frac{\partial x^*}{\partial \sigma} < 0$ whenever

$$\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D > 0.$$

Under Assumption A.1, this holds on state sets where the signed skew and dollar state do not overturn ϕ_σ .

Proof. Chain rule with $x^* = x^*(\kappa)$ and $\kappa = \kappa(\sigma, RR, D, T)$ gives the stated derivative. Since $x_k < 0$, the sign follows from the bracketed load. $\square$

[P2: Skew Amplification of Volatility Pass-Through] The cross-partial is

$$\frac{\partial^2 x^*}{\partial \sigma \partial RR} = x_{kk} \left(\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D \right) \left(\phi_{RR} + \psi_{\sigma RR} \sigma \right) + x_k \psi_{\sigma RR}.$$

A *sufficient* condition for $\frac{\partial^2 x^*}{\partial \sigma \partial RR} < 0$ at a given state (σ, RR, D) is

$$\psi_{\sigma RR} > \frac{2}{b + \kappa} \left(\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D \right) \left(\phi_{RR} + \psi_{\sigma RR} \sigma \right). \quad (8)$$

In particular, at a neutral baseline ($RR = 0$) and modest σ ,

$$\psi_{\sigma RR} > \frac{2}{b + \kappa} \phi_\sigma \phi_{RR} \implies \frac{\partial^2 x^*}{\partial \sigma \partial RR} < 0. \quad (9)$$

Proof. Differentiate $\partial x^* / \partial \sigma = x_k \partial \kappa / \partial \sigma$ w.r.t. RR :

$$\frac{\partial^2 x^*}{\partial \sigma \partial RR} = x_{kk} \left(\frac{\partial \kappa}{\partial \sigma} \right) \left(\frac{\partial \kappa}{\partial RR} \right) + x_k \left(\frac{\partial^2 \kappa}{\partial \sigma \partial RR} \right) = x_{kk}(\cdots)(\cdots) + x_k \psi_{\sigma RR}.$$

Since $x_{kk} > 0$ and $x_k < 0$, the sign is ambiguous in general. The inequality (8) is equivalent to requiring the (negative) interaction term $x_k \psi_{\sigma RR}$ to dominate the (positive) convexity term $x_{kk}(\cdot)(\cdot)$. Using $\frac{x_{kk}}{|x_k|} = \frac{2}{b + \kappa}$ yields (8); setting $RR = 0$ and evaluating at a baseline gives (9). $\square$

[P3: Dollar-State Amplification of Volatility Pass-Through] The cross-partial is

$$\frac{\partial^2 x^*}{\partial \sigma \partial D} = x_{kk} \left(\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D \right) \left(\Lambda'(D) + \psi_{\sigma D} \sigma \right) + x_k \psi_{\sigma D}.$$

A *sufficient* condition for $\frac{\partial^2 x^*}{\partial \sigma \partial D} < 0$ at a given state (σ, RR, D) is

$$\psi_{\sigma D} > \frac{2}{b + \kappa} \left(\phi_\sigma + \psi_{\sigma RR} RR + \psi_{\sigma D} D \right) \left(\Lambda'(D) + \psi_{\sigma D} \sigma \right). \quad (10)$$

In particular, at a baseline with moderate σ ,

$$\psi_{\sigma D} > \frac{2}{b + \kappa} \phi_\sigma \Lambda'(D) \implies \frac{\partial^2 x^*}{\partial \sigma \partial D} < 0. \quad (11)$$

Proof. Identical steps to Proposition A.1, replacing RR by D and using $\partial^2 \kappa / \partial \sigma \partial D = \psi_{\sigma D}$. $\square$

[P4: Term-Structure Attenuation] Let $\phi_T > 0$ and suppose the state loads $\phi_\sigma(T), \phi_{RR}(T), \psi_{\sigma RR}(T), \psi_{\sigma D}(T)$ are weakly decreasing in T (front-loaded capital/margin/PFE). Then the absolute volatil-

ity pass-through satisfies

$$\left| \frac{\partial x^*}{\partial \sigma} \right| = \frac{ab}{(b + \kappa)^2} \left(\phi_\sigma(T) + \psi_{\sigma RR}(T) RR + \psi_{\sigma D}(T) D \right),$$

which weakly *declines* with T due to (i) the explicit T -dependence of the loads and (ii) the $(b + \kappa)^{-2}$ factor when κ itself falls with T via ϕ_T/T .

Proof. Immediate from Proposition A.1, the monotonicity of the loads in T , and $\partial \kappa / \partial T = -\phi_T/T^2 + (\text{terms from load decay in } T)$. Both channels reduce $|(b + \kappa)^{-2} \cdot (\text{effective vol load})|$ as T rises. $\square$

A.2 Option-Market Microfoundations (Sketch)

Delta-hedged P&L over a short horizon Δt is $\Delta \Pi \approx \frac{1}{2} \Gamma (\Delta S)^2 + \text{vega} \cdot \Delta \sigma$. A short gamma/vega desk loses when realized or implied vol rises, with losses concentrated in the costly wing when RR is large and of the adverse sign. Under FRTB (vega/curvature), SA-CCR (RC+PFE), and CCP IM/VM, these exposures scale capital and margin; the interaction of vol with (i) skew and (ii) dollar tightness rationalizes $\psi_{\sigma RR}, \psi_{\sigma D} \geq 0$. These map into κ via KVA/MVA and funding/liquidity constraints, generating the state dependence used above.

A.3 From Valuation Adjustments to the Slope κ (Sketch)

Desks internalize costs through adjustments:

$$\kappa \approx \frac{\partial \text{KVA}}{\partial q} + \frac{\partial \text{MVA}}{\partial q} + (\text{hedging slippage}),$$

with $\partial \text{KVA} / \partial q$ increasing in FRTB sensitivities and SA-CCR EAD, and $\partial \text{MVA} / \partial q$ increasing in IM (and liquidity for VM). Procyclical multipliers make the interaction loads $\psi_{\sigma RR}, \psi_{\sigma D}$ economically natural.

A.4 Synthesis

Under Assumption A.1, Proposition A.1 delivers $\partial x^* / \partial \sigma < 0$ on economically relevant states; Propositions A.1–A.1 give explicit *sufficient conditions* (8)–(10) under which volatility pass-through is *amplified* by skew or dollar tightness (negative cross-partials); and Proposition A.1 formalizes tenor attenuation.